Economic Inflection Points in Space Manufacturing: Monte Carlo Analysis of In-Situ Resource Utilization vs. Earth Launch for Large-Scale Space Infrastructure

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Abstract

We present a parametric cost model comparing Earth-launch and in-situ resource utilization (ISRU) pathways for serial production of 1,850 kg structural modules, incorporating NPV discounting, pathway-specific schedules, and Wright learning curves. A 10,000-run Monte Carlo sampling thirteen stochastic parameters—including log-normal ISRU capital (median \$65B, $\sigma_{\ln} = 0.70$, calibrated to Flyvbjerg megaproject data) and stochastic launch cost floor ($U[\$100, \$400]$/kg)—via a 3D Gaussian copula ($\rho_{p,K} = 0.3$, $\rho_{K,\dot{n}} = 0.5$) finds crossover within 40,000 units in 68% of scenarios at $r = 5\%$ (79% at 3%, 54% at 8%), with conditional median ∼5,000 units. Rank-regression variance decomposition confirms that Earth learning rate and ISRU capital together explain ∼70% of output variance. A piecewise learning plateau model (exponent moderation at $n > 500$) confirms that results do not depend on Wright curve extrapolation beyond empirically validated volumes. The Earth pathway is validated against Iridium NEXT production data (81 satellites, implied $\mathrm{LR}_E \approx 0.85$). Over 30 robustness tests confirm that crossover is preserved under diverse assumptions, though not guaranteed: three factors

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[1] This research was conducted using an AI-assisted methodology. Claude (Anthropic) was used for literature synthesis, editorial review, and peer review simulation. The Monte Carlo simulation code was written and validated by the human author in Python using NumPy and SciPy. All quantitative results were produced by deterministic and stochastic simulation code; parameter ranges were informed by published literature and engineering analogy. No AI-generated numerical outputs were used without independent verification against the simulation code.

can prevent it—vitamin component costs above ∼$50,000/kg, discount rates above ∼20%, or technical success probability below ∼52–93%. For revenue-generating infrastructure, the opportunity cost of ISRU's deployment delay may offset savings at revenue rates above ∼$0.9M/unit/yr, suggesting the ISRU advantage is strongest for non-revenue infrastructure.

Keywords: in-situ resource utilization, space manufacturing, launch cost, Monte Carlo simulation, learning curves, net present value, space economics

1. Introduction

Any program of large-scale space infrastructure construction—whether solar power satellites, orbital habitats, or distributed energy collection architectures—eventually reaches a decision point that no amount of launch cost reduction can fully avoid under Earth-supplied logistics: at what scale does it become cheaper to manufacture components in space rather than to build them on Earth and haul them up through the gravity well? This question has been posed qualitatively since at least the 1970s [27, 28], and it has recently gained renewed urgency as launch costs decline precipitously [19], lunar ISRU programs advance under the Artemis architecture [30], and private ventures pursue asteroid resource extraction [15, 1].

The intuition is rooted in a cost asymmetry that launch cost reduction cannot fully eliminate. Reusable vehicles have brought per-kilogram costs down by roughly two orders of magnitude, from $20,000/kg to projections below $500/kg for Starship-class vehicles. But per-kilogram launch costs decline more slowly than manufacturing costs: even under aggressive launch learning ($\mathrm{LR}_L = 0.90$), the propellant and operations component (∼$200/kg) constitutes an assumed operational asymptote under Earth-supplied logistics that the learnable component cannot breach. ISRU inverts this cost structure: large fixed capital costs up front, but each additional unit produced from local materials costs only energy and labor. The fixed costs divide across a growing denominator; marginal costs decline with experience.

Despite the intuitive appeal of this argument, a quantitative gap persists in the literature. Existing ISRU economic analyses are overwhelmingly mission-specific: they evaluate the cost-effectiveness of producing lunar oxygen for propellant [30], extracting water ice for life support [39], or mining asteroidal platinum-group metals for terrestrial markets [31, 15]. We are not aware of prior work that combines schedule-aware NPV crossover anal-

ysis with systematic uncertainty characterization for generic manufactured products.

A second gap concerns the treatment of manufacturing learning. The Wright learning curve [37], well established in aerospace production economics [36], predicts that unit manufacturing cost decreases as a power function of cumulative production volume. This effect has been extensively documented for Earth-based aerospace systems [2, 25] but we are not aware of it being combined with NPV timing analysis in a comparative Earth-versus-ISRU cost model for generic structural units. The omission matters because the two pathways exhibit fundamentally different learning dynamics: Earth manufacturing benefits from learning in fabrication but exhibits limited learning in launch (baseline $\mathrm{LR}_L = 0.97$; we sweep from 90% to 100% in §4.2), while ISRU benefits from learning in both extraction and fabrication, albeit from a higher initial cost base and with a slower initial ramp-up. The long-run consequences of this asymmetry are not obvious without a quantitative model.

This paper makes three contributions. First, we present a parametric cost model with net present value discounting and pathway-specific delivery schedules that expresses the total cost of producing N units via the Earth-launch pathway and the ISRU pathway as explicit functions of production volume, with learning curves, an integrated production ramp-up schedule, ISRU mass penalty and transport costs, and financing costs applied to the appropriate components (§3). Second, we embed this model in a 10,000-run Monte Carlo framework with correlated sampling that propagates uncertainty in thirteen parameters—including the ISRU operational cost floor and production rate—at each of several fixed discount rates, yielding probability distributions, convergence analysis, and global sensitivity rankings rather than point estimates (§4). Third, we propose a phased hybrid transition strategy and discuss its implications for near-term space policy (§5).

2. Related Work

The economic case for ISRU has been examined primarily for mission-specific applications. Sanders and Larson [30] found lunar oxygen extraction cost-effective after approximately five crewed missions, but their model does not generalize to manufactured goods. Crawford [8] catalogued lunar mineralogical inventories; Cilliers et al. [7] examined regolith processing pathways; Kornuta et al. [22] developed a commercial lunar propellant architecture.

The MOXIE experiment [16] provided empirical validation of in-situ oxygen production on Mars, though at sub-manufacturing scales. Sowers [32, 33] presented NPV-based business cases for lunar ice mining and cislunar economics. Metzger et al. [24] proposed a bootstrapping approach in which a seed factory progressively builds its own capacity. Asteroid mining NPV frameworks have been developed by Sonter [31], Elvis [15], and Andrews et al. [1]; Ishimatsu et al. [17] modeled Earth–Moon–Mars logistics as a multicommodity network flow. These analyses focus on resource extraction rather than structural manufacturing. Lunar additive manufacturing demonstrations [35, 6] provide empirical grounding for the ISRU manufacturing pathway assumed here. O'Neill [27, 28] argued on energetic grounds for lunar material delivery to orbit, but lacked a parametric cost model and did not incorporate learning curves. The real options framework [10] extends NPV for irreversible investments under uncertainty; Saleh et al. [29] applied flexibility analysis to spacecraft, and de Weck et al. [9] demonstrated staged deployment value. Organizational forgetting [5, 34] has been documented in aerospace production lines.

Jones [19, 20, 21] documented a roughly two-orders-of-magnitude reduction in launch cost per kilogram over six decades. A critical distinction is between vehicle manufacturing cost (which follows a learning curve) and per-kilogram delivery cost (dominated by propellant and operations, largely independent of cumulative launches). Falcon 9 reuse economics [38] confirm that per-launch cost reductions arise from fixed-cost amortization at high flight rates, not production learning. Wertz [36] estimated launch vehicle production learning rates of 85–90%; the NASA Cost Estimating Handbook [26] provides standardized parametric methodologies that inform our parameter selection. Our model adopts a two-component launch cost: a propellant operational asymptote plus a learnable component ($\mathrm{LR}_L = 0.97$); we sweep LR_L from 0.90 to 1.00 in §4.2.

The Wright learning curve [37]—in which unit cost decreases as a power function of cumulative production—is well established in aerospace economics [11, 2, 25]. The canonical form is

$$C(n) = C(1) \cdot n^b, \quad b = \frac{\ln(\mathrm{LR})}{\ln 2}, \tag{1}$$

where $C(n)$ is the cost of the nth unit, $C(1)$ is the first-unit cost, and LR is the learning rate. Typical aerospace learning rates range from 0.80 to 0.95 [36]. Baumers et al. [4] documented rates of 0.85–0.92 for metal additive

manufacturing, supporting our ISRU baseline ($\mathrm{LR}_I = 0.90$). We are not aware of prior work applying the Wright model to a comparative Earth-versus-ISRU manufacturing analysis with NPV timing.

3. Model Description

Decision problem.. The primary comparison metric is the NPV of cumulative cost to deliver the first N units (Eq. 16), where each pathway delivers according to its own schedule. This answers: "Given a program requirement of N structural modules, which pathway has lower total present-value cost?" Section 5 extends this to a utility-maximizing framing (Eq. 23) that accounts for revenue per unit and the opportunity cost of ISRU's deployment delay.

We model the total cost of producing N identical structural modules (hereafter "units") via two pathways: an Earth-launch pathway and an ISRU pathway. Each unit has a reference mass $m = 1{,}850$ kg, representative of structural modules in the 1,000–5,000 kg class typical of solar collector or orbital habitat components, though the model is parameterizable to other mass ranges. We describe each pathway's cost model, then the Monte Carlo framework used to propagate parameter uncertainty.

Reference orbit.. Throughout this paper, "operational orbit" refers to geostationary Earth orbit (GEO), consistent with space solar power and communications relay architectures. The launch cost p_{launch} represents delivery from Earth surface to GEO; the transport cost $p_{\mathrm{transport}}$ represents delivery from the lunar surface to GEO via low-energy trajectory ($\sim$6 km/s Δv). For other destinations (LEO, NRHO, cislunar), both cost terms would differ; the model is parameterizable to any orbit.

Cost basis normalization.. All launch costs in this paper are quoted for delivery to GEO. Published Starship cost projections (\$200–500/kg) typically refer to LEO; delivery to GEO via an upper stage or electric tug adds a factor of $\sim$2–3$\times$, so \$500/kg to LEO corresponds to approximately \$1,000–1,500/kg to GEO, consistent with our baseline of \$1,000/kg. The operational asymptote p_{fuel} is now sampled stochastically: $p_{\mathrm{fuel}} \sim U[\$100, \$400]$/kg (baseline \$200/kg). This range is grounded in a bottom-up decomposition of the GEO delivery cost chain:

1. *Propellant to LEO*: LOX at $\sim$\$0.20/kg and CH_4 at $\sim$\$0.40/kg, with a propellant-to-payload mass ratio of $\sim$25:1 for Starship-class vehicles, yields $\sim$\$5–8/kg of payload.

2. *LEO-to-GEO transfer*: $\Delta v \approx 3.9$ km/s (Hohmann) to 4.2 km/s (with plane change); a chemical tug costs ∼\$80–120/kg transferred (propellant mass fraction plus amortized tug operations), or ∼\$30–60/kg via slower solar electric propulsion.
3. *Ground operations*: Pad turnaround, range fees, and GSE amortization at high cadence (>50 flights/yr) yield ∼\$20–50/kg.

Summing these components gives a total floor of ∼\$105–178/kg to GEO, with the \$200/kg baseline representing the central estimate including operational margin. The $U[\$100, \$400]$ range brackets the bottom-up floor at the low end and allows for higher costs under less optimistic ground operations or chemical tug assumptions at the high end.

3.1. Earth-launch pathway

The cost of the nth unit produced on Earth and launched to the operational orbit is

$$C_{\text{Earth}}(n) = C_{\text{mfg}}(n) + C_{\text{launch}}(n), \tag{2}$$

where $C_{\text{mfg}}(n)$ is the terrestrial manufacturing cost and $C_{\text{launch}}(n)$ is the launch cost. Manufacturing cost follows a two-component model separating non-learnable material costs from learnable labor and overhead:

$$C_{\text{mfg}}(n) = C_{\text{mat}} + C^{(1)}_{\text{labor}} \cdot n^{b_E}, \quad b_E = \frac{\ln(\text{LR}_E)}{\ln 2}, \tag{3}$$

where $C_{\text{mat}} = \$1\text{M}$ is the per-unit material cost in constant 2024 real dollars (aerospace-grade structural materials at ∼\$540/kg × 1,850 kg ≈ \$1M), which does not decline with cumulative production, and $C^{(1)}_{\text{labor}} = \$74\text{M}$ is the first-unit labor, overhead, and NRE amortization cost, which follows the Wright learning curve with Earth learning rate LR_E. The first-unit total is $C^{(1)}_{\text{mfg}} = C_{\text{mat}} + C^{(1)}_{\text{labor}} = \$1\text{M} + \$74\text{M} = \$75\text{M}$. The material cost serves as a natural manufacturing cost floor—more physically motivated than an arbitrary floor parameter—because no amount of production learning eliminates the cost of raw materials. At $n = 10{,}000$, the labor component has declined to ∼\$7.2M while the material cost remains \$1M, yielding a total of ∼\$8.2M—compared to ∼\$8.2M under the single Wright curve. The two-component model produces nearly identical crossover results at baseline but provides a clearer physical interpretation: the floor is material cost, not an arbitrary truncation. For backward compatibility, we retain a manufacturing

cost floor parameter $C_{\text{mfg}}^{\text{floor}}$ that can be applied on top of the two-component model; the baseline uses $C_{\text{mfg}}^{\text{floor}} = 0$.

$$C_{\text{mfg}}(n) = \max\Big(C_{\text{mat}} + C_{\text{labor}}^{(1)} \cdot n^{\,b_E},\ C_{\text{mfg}}^{\text{floor}}\Big). \tag{4}$$

The baseline launch cost is treated as an exogenous parameter with no endogenous learning:

$$C_{\text{launch}}(n) = m \cdot p_{\text{launch}}, \tag{5}$$

i.e., a constant \$1,000/kg × 1,850 kg = \$1.85M per unit. This is appropriate when the program's launch demand is a small fraction of total market demand, so launch cost reductions are driven by industry-wide experience rather than the program itself.

As a sensitivity variant, we test a two-component model with program-indexed learning:

$$C_{\text{launch}}^{\text{learn}}(n) = m\left[p_{\text{fuel}} + p_{\text{ops}} \cdot n^{\,b_L}\right], \quad b_L = \frac{\ln(\text{LR}_L)}{\ln 2}, \tag{6}$$

where $p_{\text{fuel}} = \$200/\text{kg}$ is an assumed operational asymptote under Earth-supplied propellant logistics (propellant plus minimal GEO delivery operations) and $p_{\text{ops}} = \$800/\text{kg}$ is the learnable component, declining at $\text{LR}_L = 0.97$ per doubling. This two-component model is reported in §4.2 (Table 5); it shifts the crossover by +272 units (+7%) relative to the no-learning baseline, confirming that the model's conclusions are insensitive to the launch learning assumption. **Indexing convention.** The learning index n counts cumulative program units. At the assumed program scale (∼4,100–10,000 units), the program would constitute a substantial fraction of global launch demand, making program-indexed learning a reasonable proxy for market-indexed learning. Nevertheless, the no-learning baseline is the primary formulation because it avoids conflating endogenous program scale effects with exogenous market trends.

The cumulative cost of producing N units via the Earth pathway is therefore

$$\Sigma_{\text{Earth}}(N) = \sum_{n=1}^{N}\left[\max\Big(C_{\text{mat}} + C_{\text{labor}}^{(1)} \cdot n^{\,b_E},\ C_{\text{mfg}}^{\text{floor}}\Big) + m \cdot p_{\text{launch}}\right]. \tag{7}$$

When the launch learning sensitivity variant (Eq. 6) is active, $m \cdot p_{\text{launch}}$ is replaced by the two-component expression.

3.2. ISRU pathway

3.2.1. Production schedule and ramp-up

A critical distinction between the two pathways is their delivery timing. Earth manufacturing can begin immediately upon program start, requiring no new infrastructure beyond existing terrestrial supply chains. The ISRU pathway, by contrast, requires construction and commissioning of an extraterrestrial facility before production can begin, followed by a logistic ramp-up to full capacity.

Earth delivery schedule.. The Earth pathway delivers units at a constant rate from $t = 0$:

$$t_{n,E} = \frac{n}{\dot{n}_{\max}}, \tag{8}$$

where $\dot{n}_{\max}$ is the full-capacity production rate (units/year). This assumes pre-existing Earth manufacturing capacity, reasonable given the maturity of terrestrial aerospace production; we test a logistic Earth ramp-up ($t_{0,E} \in \{1, 2\}$ yr) in §4.5.

ISRU delivery schedule.. The ISRU facility's instantaneous production rate follows a logistic ramp-up function:

$$\dot{n}(t) = \dot{n}_{\max} \cdot S(t), \quad S(t) = \frac{1}{1 + e^{-k(t-t_0)}}, \tag{9}$$

where t_0 is the ramp-up midpoint (years from program start) and $k = 2.0$ (fixed) controls steepness. Integrating the production rate yields the cumulative production function:

$$N(t) = \frac{\dot{n}_{\max}}{k} \left[\ln\!\left(1 + e^{k(t-t_0)}\right) - \ln 2 \right], \tag{10}$$

The constant $-\ln 2$ ensures $N(t_0) = 0$. A piecewise formulation enforces $\dot{n}(t) = 0$ for $t < t_c$ ($t_c = t_0 - 1$ by default); at $t = t_0$ the rate is half-maximal, exceeding 90% capacity within $\sim$1 year (Figure 7). The closed-form inverse gives the calendar time of the nth ISRU unit:

$$t_{n,I} = t_0 + \frac{1}{k} \ln\!\left(2\, e^{nk/\dot{n}_{\max,\mathrm{eff}}} - 1\right). \tag{11}$$

Schedule model verification.. Substituting $n = 1$ into Eq. 11 yields $t_{1,I} = t_0 + (1/k)\ln(2e^{k/\dot{n}_{\max}} - 1) \approx t_0$, confirming that the first unit is produced near the ramp-up midpoint, with $S(t_0) = 0.50$. The piecewise enforcement ($\dot{n} = 0$ for $t < t_c$) affects only the exponentially small tail before t_c; $N(t_c)$ is negligible ($\sim 10^{-3}$ units). The counting convention starts from $N(t_0) = 0$, not from facility groundbreaking.

The effective ISRU production rate accounts for facility availability:

$$\dot{n}_{\max,\text{eff}} = A \cdot \dot{n}_{\max}, \tag{12}$$

where $A \in [0, 1]$ is the fraction of calendar time the facility is operational. The deterministic baseline uses $A = 1.0$ (backward-compatible); the Monte Carlo samples $A \sim U[0.70, 0.95]$.

Timing gap.. The separation between these schedules is substantial. Table 1 shows that when Earth has delivered its 1,000th unit (at $t = 2$ yr), the ISRU pathway has not yet produced any units (ISRU's 1,000th unit arrives at $t \approx 7.3$ yr). This gap has a counterintuitive NPV consequence: because Earth costs are incurred earlier, they are discounted *less* (i.e., carry higher present value), making the Earth pathway more expensive in NPV terms than its nominal costs suggest. This effect partially offsets the ISRU pathway's heavy upfront capital burden.

Table 1: Pathway-specific delivery schedules. The gap column shows the additional time ISRU requires relative to Earth delivery of the same unit number.

Unit n	$t_{n,E}$ (yr)	$t_{n,I}$ (yr)	$S(t_{n,I})$	Gap (yr)
1	0.002	5.00	0.50	+5.00
10	0.02	5.04	0.52	+5.02
100	0.20	5.34	0.66	+5.14
500	1.00	6.31	0.93	+5.31
1,000	2.00	7.34	0.99	+5.34
5,000	10.00	15.35	1.00	+5.35
10,000	20.00	25.35	1.00	+5.35

3.2.2. Cost model

For illustrative comparison of per-unit costs, the ISRU cost of the nth unit can be expressed as an amortized capital component plus an operational

component:

$$C_{\mathrm{ISRU}}(n) = \frac{K}{N_{\mathrm{total}}} + C_{\mathrm{ops}}(n), \tag{13}$$

where K is the total capital investment required to establish the ISRU infrastructure, and N_{total} is a planning horizon over which capital is amortized for display purposes. We set $N_{\mathrm{total}} = 10{,}000$ throughout; this choice affects the per-unit cost visualization (Figure 3) but does not enter the crossover calculation, which uses the cumulative formulation below.

The operational cost follows a Wright learning curve with a cost floor C_{floor} representing irreducible per-unit expenses (energy, consumables, remote operations overhead), scaled by a mass penalty factor α and augmented by a transport cost for delivering ISRU-manufactured units to the operational orbit:

$$C_{\mathrm{ops}}(n) = \alpha \left[C_{\mathrm{floor}} + \left(C_{\mathrm{ops}}^{(1)} - C_{\mathrm{floor}} \right) \cdot n^{\,b_I} \right] + m \cdot p_{\mathrm{transport}} \cdot \alpha, \tag{14}$$

where $b_I = \ln(\mathrm{LR}_I)/\ln 2$ with ISRU learning rate LR_I, C_{floor} represents the asymptotic minimum operational cost per unit (energy, consumables, remote operations overhead), $\alpha \geq 1$ is a delivered-mass penalty: an ISRU-manufactured unit with $\alpha = 1.3$ has 30% more mass than the reference design (thicker structural margins, lower-purity alloys) and therefore requires proportionally more processing and transport—both operational cost and transport cost scale with α because both are proportional to the mass actually processed and delivered—and $p_{\mathrm{transport}}$ (\$/kg) is the cost of transporting an ISRU-manufactured unit from the production site to the operational orbit. The cost floor is sampled as $C_{\mathrm{floor}} \sim U[\$0.3\mathrm{M}, \$2.0\mathrm{M}]$ in the Monte Carlo; when C_{floor} exceeds the Earth pathway's per-unit launch cost floor ($m \cdot p_{\mathrm{launch}} = \$1.85\mathrm{M}$ at baseline), the per-unit cost advantage of ISRU is reduced, potentially preventing crossover within the planning horizon. This scenario is captured in the non-convergence statistics (§4.3).

The ramp-up effect is captured purely through timing: early units are produced later and therefore discounted more heavily in the NPV calculation (Eq. 11). The cumulative cost of producing N units via the ISRU pathway is

$$\Sigma_{\mathrm{ISRU}}(N) = K + \sum_{n=1}^{N} C_{\mathrm{ops}}(n). \tag{15}$$

Note that the full capital cost K is incurred upfront; the amortized per-unit formulation (Eq. 13) is used only for visualization.

3.2.3. Crossover and NPV formulation

The undiscounted crossover N_0^* is the smallest N with $\Sigma_{\text{ISRU}}(N) \leq \Sigma_{\text{Earth}}(N)$. Because the ISRU pathway concentrates expenditure early while Earth distributes costs over time, each pathway's costs must be discounted according to its own delivery schedule. We define the NPV crossover N^* as the smallest N satisfying

$$K + \sum_{n=1}^{N} \frac{C_{\text{ops}}(n)}{(1+r)^{t_{n,I}}} \;\leq\; \sum_{n=1}^{N} \frac{C_{\text{Earth}}(n)}{(1+r)^{t_{n,E}}}, \tag{16}$$

where r is the real discount rate (all costs in constant 2024 USD). In the baseline formulation, K is incurred at $t = 0$; in the phased formulation (§4.4), K is spread over five annual tranches. Because Earth costs are incurred earlier ($t_{n,E} < t_{n,I}$), they carry higher present value, making the Earth pathway more expensive in NPV terms; this partially offsets the ISRU pathway's upfront capital burden. When $r = 0$, Eq. 16 reduces to the undiscounted comparison.

3.2.4. Earth-sourced component fraction

Even passive structural modules require Earth-sourced fasteners, seals, sensors, and coatings. We model this via a vitamin fraction f_v: a fraction f_v of each unit's mass must be launched from Earth, with its own manufacturing cost c_{vit} (\$/kg). The ISRU operational cost becomes

$$C_{\text{ops}}^{\text{vit}}(n) = (1 - f_v) \cdot C_{\text{ops}}(n) + f_v \cdot m \cdot \big(p_{\text{launch,eff}}(n) + c_{\text{vit}}\big), \tag{17}$$

where $p_{\text{launch,eff}}(n)$ includes launch learning when active, and $c_{\text{vit}} = \$10{,}000/\text{kg}$ (space-qualified electronics). The baseline uses $f_v = 0.05$ (5% of unit mass Earth-sourced). This vitamin fraction is included in the baseline because it drives the permanent/transient crossover distinction (§4.3): at large n, the vitamin component's asymptotic cost ($f_v \cdot m \cdot (p_{\text{fuel}} + c_{\text{vit}}) \approx \$0.94\text{M}$) raises the ISRU floor above the Earth pathway's asymptotic cost, making most crossovers transient (finite-horizon amortization effects). This is a conservative modeling choice; if vitamin components can eventually be ISRU-sourced, all crossovers become permanent. The $f_v = 0$ case (fully ISRU-derived, an optimistic bound) is reported as a sensitivity variant. We test $f_v \in \{0, 0.05, 0.10, 0.15, 0.20\}$ and $c_{\text{vit}} \in \{\$5{,}000, \$10{,}000, \$50{,}000\}/\text{kg}$ in §4.2; crossover is preserved up to $f_v = 15\%$, confirming that ISRU need not produce all components to achieve economic advantage.

3.3. Monte Carlo framework

To characterize the sensitivity of N^* to parameter uncertainty, we embed the NPV cost model in a Monte Carlo simulation. Each of 10,000 runs samples thirteen parameter values from the distributions specified in Table 2 and computes the NPV crossover point via Eq. 16. We define "achieving crossover within horizon H" as the event $N^* \leq H$; throughout this paper, $H = 40{,}000$ unless otherwise stated. Critically, the discount rate r is *not* treated as a stochastic parameter. Instead, the full Monte Carlo ensemble is run independently at each of several fixed discount rates ($r \in \{3\%, 5\%, 8\%\}$). This separation is motivated by the observation that the discount rate reflects the decision-maker's time preference and financing structure rather than technological uncertainty; conflating it with stochastic cost parameters obscures the distinct roles of economic policy and engineering risk [3]. We report the median, interquartile range, and 10th/90th percentiles of the resulting crossover distribution at each rate, with 95% bootstrap confidence intervals computed from 5,000 bootstrap resamples.

Learning rate parameters are sampled from clipped normal distributions (Table 2). Three parameters—launch cost, ISRU capital, and production rate—are correlated via a 3D Gaussian copula with correlation matrix

$$R = \begin{pmatrix} 1 & 0.3 & 0 \\ 0.3 & 1 & 0.5 \\ 0 & 0.5 & 1 \end{pmatrix}, \tag{18}$$

where rows/columns correspond to $(p_{\text{launch}}, K, \dot{n}_{\max})$. The launch–capital correlation ($\rho_{p,K} = 0.3$) captures common factors (regulatory environment, technology maturity). The capital–production rate correlation ($\rho_{K,\dot{n}} = 0.5$) reflects that higher-capacity facilities require larger investment, eliminating implausible corners (e.g., \$100B capital with 250 units/yr); independence ($\rho_{K,\dot{n}} = 0$) is tested as a sensitivity variant (Appendix Appendix A). All other parameters are sampled independently. Copula sensitivity tests ($\rho_{p,K} \in \{0, 0.3, 0.6\}$, conditional median variation <200 units) are in Appendix Appendix A. The sampled p_{launch} is decomposed per Eq. 6; the propellant floor p_{fuel} is now sampled stochastically ($U[\$100, \$400]$/kg) to reflect uncertainty in the GEO delivery cost asymptote. For global sensitivity analysis, we compute Spearman rank correlations and partial rank correlation coefficients between each input parameter and N^*.

Table 2: Monte Carlo parameter distributions. Launch cost, ISRU capital, and production rate are correlated via a 3D Gaussian copula ($\rho_{p,K} = 0.3$, $\rho_{K,\dot{n}} = 0.5$); all other parameters are sampled independently. ISRU capital follows a log-normal distribution calibrated to Flyvbjerg's megaproject reference class (P90/P50 $\approx 2.5\times$). The Monte Carlo is run at fixed discount rates $r \in \{3\%, 5\%, 8\%\}$. Achieving crossover is defined as $N^* \leq H$ where $H = 40{,}000$. Baseline values are rounded for exposition; the simulation uses exact values as specified in the code repository.

Parameter	Baseline	Distribution	Range
Stochastic parameters (13)			
Launch cost p_{launch} (\$/kg)	1,000	Uniform†	[500, 2,000]
ISRU capital K (\$B)	50	Log-normal†	med. 65, $\sigma_{\ln}$=0.70, clip [20, 200]
Earth learning rate LR_E	0.85	$\mathcal{N}(0.85, 0.03)$	[0.75, 0.95]
ISRU learning rate LR_I	0.90	$\mathcal{N}(0.90, 0.03)$	[0.80, 0.98]
Ramp-up time t_0 (years)	5	Uniform	[3, 8]
First-unit ops. cost $C_{\text{ops}}^{(1)}$ (\$M)	5	Uniform	[2, 10]
First-unit mfg. cost $C_{\text{mfg}}^{(1)}$ (\$M)	75	Uniform	[50, 100]
Mass penalty α	1.0	Uniform	[1.0, 2.0]
Transport cost $p_{\text{transport}}$ (\$/kg)	100	Uniform	[50, 300]
Ops cost floor C_{floor} (\$M)	0.5	Uniform	[0.3, 2.0]
Production rate $\dot{n}_{\max}$ (units/yr)	500	Uniform	[250, 750]
ISRU availability A	1.0‡	Uniform	[0.70, 0.95]
Launch cost floor p_{fuel} (\$/kg)	200	Uniform	[100, 400]
Fixed parameters			
Discount rate r	0.05	Fixed per run	$\{0.03, 0.05, 0.08\}$
Unit mass m (kg)	1,850	Fixed	—
S-curve steepness k	2.0	Fixed	—
Amortization horizon N_{total}	10,000	Fixed	—
Launch–capital correlation $\rho_{p,K}$	0.3	Fixed	—
Capital–production rate corr. $\rho_{K,\dot{n}}$	0.5	Fixed	—

†Launch cost and capital correlated ($\rho_{p,K} = 0.3$); capital and prod. rate correlated ($\rho_{K,\dot{n}} = 0.5$) via 3D Gaussian co

‡Baseline value for backward compatibility; the MC samples $A \sim U[0.70, 0.95]$.

3.4. Parameter justification

The most consequential parameter values are summarized here; detailed derivations, cross-checks, and empirical analogies are provided in Appendix Appendix C.

First-unit manufacturing cost $C^{(1)}_{\text{mfg}} = \$75M$ reflects a one-off spacecraft-class structural module of 1,850 kg, sampled $U[\$50\text{M}, \$100\text{M}]$. *First-unit ISRU operational cost* $C^{(1)}_{\text{ops}} = \$5M$ is derived from the energy budget for regolith processing (∼5,000 kWh at \$100–200/kWh [7]) plus equipment wear and operations overhead, sampled $U[\$2\text{M}, \$10\text{M}]$. *ISRU capital* K (median \$65B) covers the complete vertically-integrated infrastructure chain, sampled from a log-normal distribution ($\sigma_{\ln} = 0.70$, clipped to [\$20B, \$200B]). The log-normal form reflects reference-class forecasting for first-of-kind megaprojects: Flyvbjerg [12] documents P90/P50 cost overrun ratios of 2–4× for large infrastructure; our $\sigma_{\ln} = 0.70$ yields P90/P50 $\approx 2.5\times$, at the conservative end of this range. Space-specific precedents are more severe: JWST exhibited ∼10× cost growth (\$1B initial → \$10B final), and ISS ∼3×. The uniform distribution ($U[\$30\text{B}, \$100\text{B}]$) is retained as a sensitivity variant. For context, ISS lifecycle cost is ∼\$150B and NASA estimates \$5–15B for individual lunar processing facilities [30, 26]. *Production rate* $\dot{n}_{\max} = 500$ *units/year*, sampled $U[250, 750]$, requires ∼5,000–10,000 tonnes/year of raw regolith, within proposed ISRU facility scales [30]. *Learning rates*: $\text{LR}_E \sim \mathcal{N}(0.85, 0.03)$ is calibrated from Table 3; $\text{LR}_I \sim \mathcal{N}(0.90, 0.03)$ is conservative relative to terrestrial additive manufacturing analogs (Table 3), reflecting ISRU-specific challenges (remote operations, dust contamination, communication delays). No direct empirical data for extraterrestrial manufacturing exists; the ISRU range brackets the terrestrial AM analogs with a conservative offset.

Table 3: Empirical learning rates from analogous production programs. Our ranges $\text{LR}_E \sim \mathcal{N}(0.85, 0.03)$ [0.75, 0.95] and $\text{LR}_I \sim \mathcal{N}(0.90, 0.03)$ [0.80, 0.98] encompass these empirical values.

Program class	LR range	n range	Source
Satellite buses	0.83–0.88	50–200	Wertz 2011 [36]
Solar arrays	0.80–0.85	100–1,000	Nagy et al. 2013 [25]
Aircraft structures	0.80–0.85	500+	Argote & Epple 1990 [2]
Metal additive mfg.	0.85–0.92	10–100	Baumers et al. 2016 [4]

Mass penalty $\alpha \sim U[1.0, 2.0]$ captures additional feedstock required for

structural equivalence. *Availability* $A \sim U[0.70, 0.95]$ accounts for equipment failures and limited maintenance access. *Transport cost* $p_{\text{transport}} \sim U[\$50, \$300]$/kg covers lunar surface-to-GEO delivery [17].

3.5. Assumptions and limitations

Key assumptions are listed here; detailed discussion is in Appendix Appendix D.

- *Scope*: passive structural modules only; system-level crossover depends on structural cost fraction ($\sim 1/f_c$ scaling).
- *Cash-flow timing*: pay-at-milestone for both pathways; manufacturing lead-time sensitivity (-3.6% to -6.6%) confirms conservatism (Appendix Appendix A).
- *Quality parity*: Earth and ISRU units meet identical specs; optimistic for early ISRU production.
- *Fixed unit mass*: no design evolution over the production run.
- *Single product type*: multi-product facilities would shift crossover earlier.
- *Constant real discount rate*: risk-adjusted or pathway-specific rates tested as sensitivities (§4.6).
- *Earth instant start*: tested via ramp-up robustness checks (§4.5).

4. Results

4.1. Baseline crossover

Under baseline parameter values (Table 2, including $f_v = 0.05$), the undiscounted cumulative cost curves ($r = 0$) intersect at approximately $N^* = 3{,}733$ units. Figure 1 shows these trajectories. The ISRU pathway begins with a steep initial outlay (the \$50B capital investment plus operational and transport costs during ramp-up), while the Earth pathway accumulates cost at a rate dominated by the constant launch cost per unit. The Earth curve is approximately linear at large N because the learning-curve reduction in manufacturing cost is small relative to the constant launch cost per unit (\$1.85M/unit at \$1,000/kg for a 1,850 kg unit). The ISRU curve,

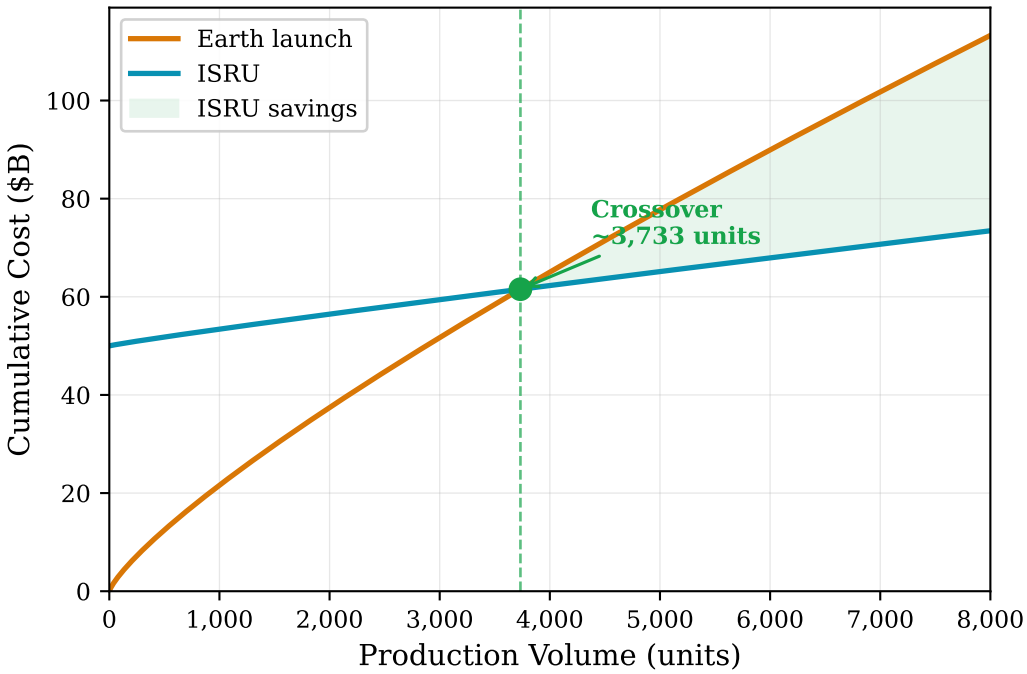

Figure 1: Undiscounted cumulative cost ($B) vs. production volume for the Earth-launch and ISRU pathways under baseline assumptions ($r = 0$, $f_v = 0.05$). The crossover occurs at approximately 3,733 units. The shaded region indicates cumulative ISRU savings.

by contrast, flattens as operational learning drives per-unit costs toward the cost floor plus the transport cost.

Applying NPV discounting at 5% with pathway-specific delivery schedules (Eq. 16) shifts the crossover to approximately $N^* = 4{,}403$ units—an 18% increase. Because Earth costs are incurred earlier and thus discounted less, they carry higher present value (see §3), partially offsetting the ISRU capital burden. Figure 2 shows cumulative cost curves at four discount rates. At $r = 10\%$, the crossover reaches ∼5,802 units.

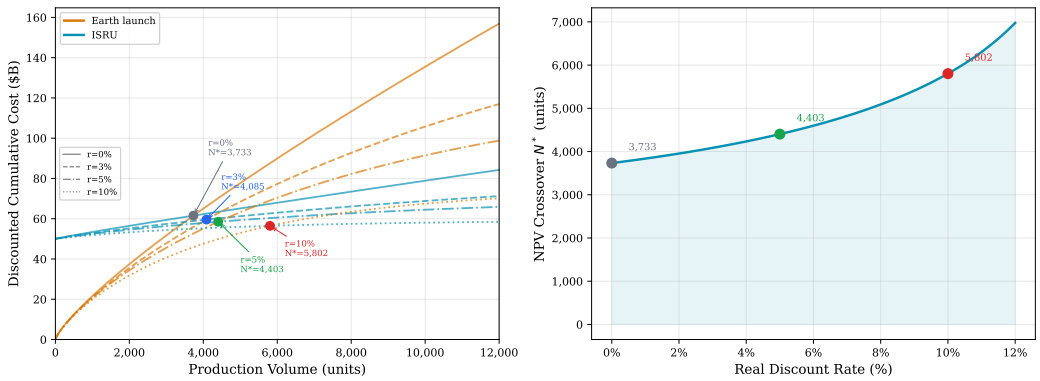

Figure 2: Left: Discounted cumulative cost curves at four discount rates with pathway-specific delivery schedules, with crossover points marked. Right: NPV crossover point as a continuous function of discount rate. The pathway-specific formulation yields lower crossover points than earlier analyses because Earth costs carry greater present-value weight.

Table 4 summarizes the crossover point under representative scenarios at both $r = 0$ and $r = 5\%$.

Table 4: Crossover points under representative scenarios at two discount rates, using pathway-specific delivery schedules.

			Undiscounted		NPV ($r = 5\%$)	
Scenario	\$/kg	K (\$B)	N^*	Time	N^*	Time
Optimistic	500	30	∼1,920	∼9 yr	∼2,016	∼9 yr
Baseline	1,000	50	∼3,733	∼12 yr	∼4,403	∼14 yr
Conservative	2,000	100	∼9,201	∼23 yr	∼23,635	∼52 yr

Figure 3 illustrates the per-unit cost dynamics. The Earth per-unit cost is bounded below by $m \cdot p_{\text{launch}}$, while ISRU declines toward $C_{\text{floor}} = \$0.5\text{M}$—

this asymptotic divergence is the structural driver of the crossover. However, as the Monte Carlo analysis shows (§4.3), crossover is achieved in 62–89% of stochastic scenarios, not in all cases; see §4.3 for the permanent/transient crossover breakdown. An Earth pathway sanity check against satellite production economics is provided in Appendix Appendix C.

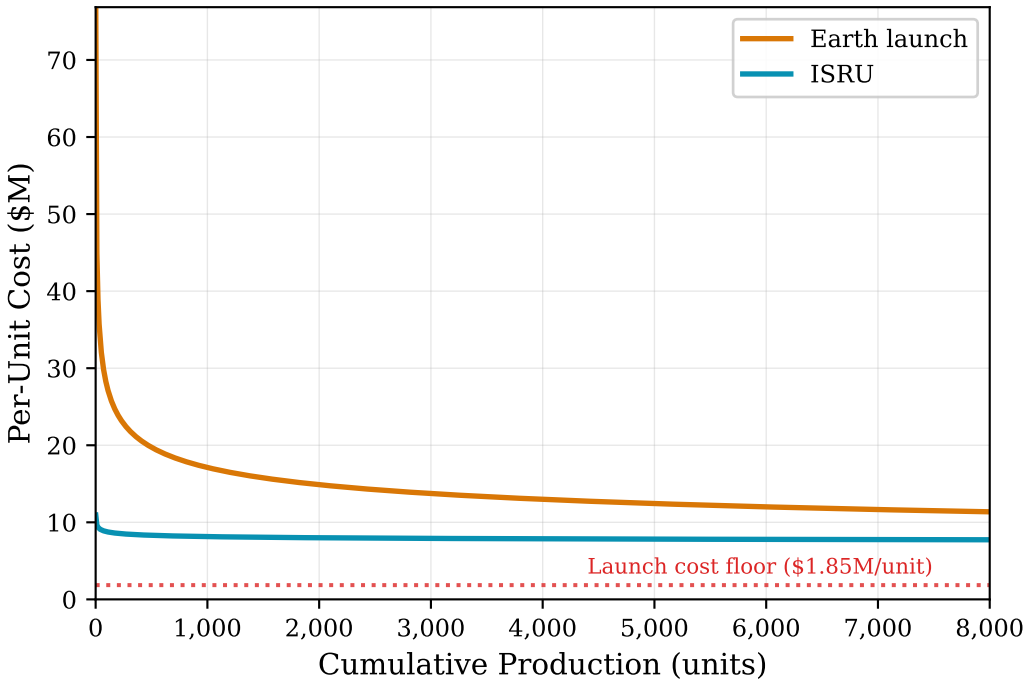

Figure 3: Per-unit cost ($M) vs. cumulative production for the Earth-launch and ISRU pathways. The Earth pathway is bounded below by the launch cost floor ($m \cdot p_{\text{launch}} =$ \$1.85M), while the ISRU pathway declines toward the operational cost floor ($C_{\text{floor}} =$ \$0.5M).

4.2. Sensitivity analysis

Figure 4 summarizes one-at-a-time sensitivity of the NPV crossover ($r = 5\%$) to nine parameters.

The Earth manufacturing learning rate remains the most influential parameter in absolute range, with a strongly asymmetric effect: a *higher* learning rate ($\text{LR}_E = 0.90$, i.e., slower learning—each doubling reduces cost by only 10%) shifts the crossover earlier because the Earth pathway retains higher costs for longer, while a *lower* rate ($\text{LR}_E = 0.80$, faster learning)

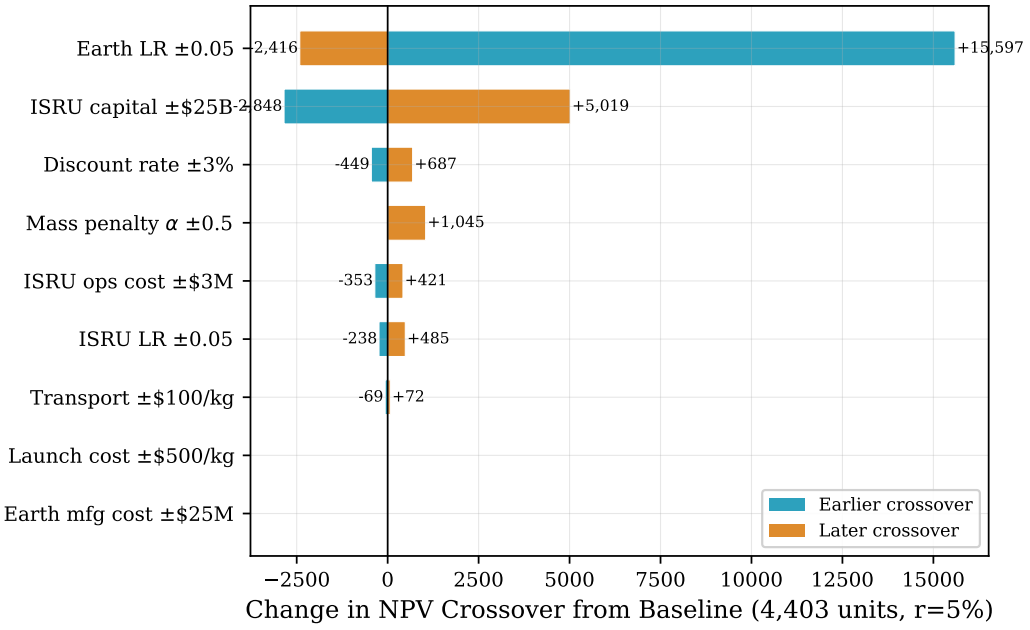

Figure 4: Tornado diagram showing the ranked sensitivity of the NPV crossover point ($r = 5\%$ baseline, pathway-specific timing) to each input parameter.

delays crossover because Earth manufacturing costs decline rapidly. ISRU capital investment is the second most influential parameter. The discount rate r remains important but its effect is moderated under pathway-specific timing because the timing gap between pathways partially offsets the discounting penalty.

The mass penalty factor α (varied from 1.0 to 2.0) shifts the crossover by approximately +440 units at $\alpha = 1.5$. Transport cost $p_{\text{transport}}$ has a smaller effect, consistent with transport being a modest fraction of total ISRU cost at baseline parameters. First-unit costs ($C^{(1)}_{\text{ops}}$, $C^{(1)}_{\text{mfg}}$) exert moderate influence. ISRU learning rate and launch cost remain relatively weak individual drivers, though their effects compound with other parameters in the Monte Carlo analysis.

The two-dimensional parameter space of launch cost versus ISRU capital is shown in the heat map of Figure 5, now computed using the NPV crossover at $r = 5\%$ with pathway-specific timing.

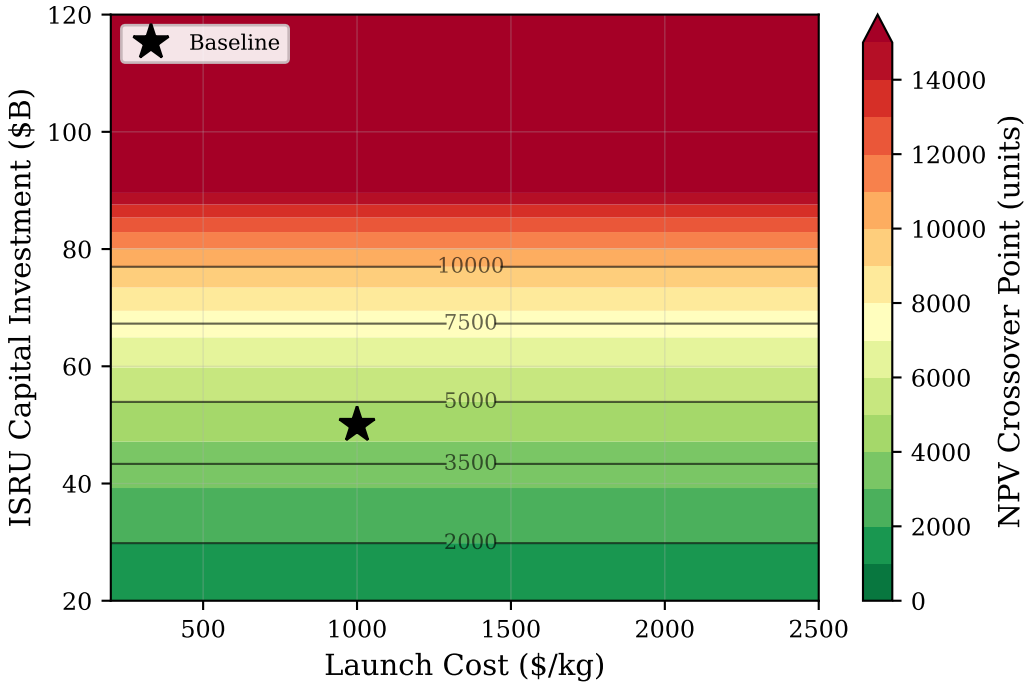

Figure 5: NPV crossover point N^* ($r = 5\%$, pathway-specific timing) as a function of launch cost ($/kg) and ISRU capital investment ($B). The star marks the baseline scenario.

Launch cost learning sweep.. The baseline model uses constant launch cost (Eq. 5). As a sensitivity variant, we test the two-component learning model (Eq. 6) with LR_L ranging from 0.90 (aggressive: 10% cost reduction per doubling) to 0.97 (moderate). Table 5 reports the results. Even at the aggressive $\mathrm{LR}_L = 0.90$—where the ops component falls to ∼\$334/kg by $n = 10{,}000$—the crossover shifts by only +315 units from baseline (+7%). The no-learning case ($\mathrm{LR}_L = 1.00$) gives $N^* = 4{,}403$, identical to baseline because the baseline launch cost already uses this as the fixed-cost formulation. The reason launch learning cannot eliminate the ISRU advantage is structural: even at $\mathrm{LR}_L = 0.90$, the fuel component (\$200/kg) constitutes an assumed operational asymptote under Earth-supplied propellant logistics. At $n = 10{,}000$, the ops component has fallen to ∼\$334/kg, but the combined launch cost (\$534/kg × 1,850 kg = \$0.99M per unit) remains above the ISRU operational cost floor. This floor asymmetry—physics-driven propellant costs versus experience-driven operational costs—is the fundamental reason that launch learning, regardless of its rate, cannot fully close the gap at high production volumes. §4.2 extends this analysis to the scenario where ISRU-produced propellant reduces the fuel floor itself.

Table 5: Launch cost learning sweep (NPV, $r = 5\%$). The fuel component (\$200/kg) is held constant; the operational component (\$800/kg) follows a Wright learning curve. Baseline: $\mathrm{LR}_L = 0.97$.

LR_L	b_L	N^*	Shift	Note
1.00	0.000	4,403	±0	No learning (= baseline)
0.99	−0.014	4,280	−123	Minimal learning
0.97	−0.044	4,403	—	Baseline
0.95	−0.074	4,512	+109	Moderate learning
0.93	−0.105	4,605	+202	
0.90	−0.152	4,718	+315	Aggressive learning

ISRU learning rate boundary scenarios.. To bound the sensitivity of the crossover to the ISRU learning rate—an assumption with limited empirical grounding (see §3.4)—we test two extreme scenarios. With *no ISRU learning* ($\mathrm{LR}_I = 1.0$, operational costs remain constant at $C_{\mathrm{ops}}^{(1)}$), the NPV crossover shifts to ∼6,082 units (+1,679 from baseline). With *very slow learning* ($\mathrm{LR}_I = 0.98$), the crossover shifts to ∼5,449 units (+1,046). In both cases the crossover still occurs, demonstrating that the ISRU economic

advantage does not depend on optimistic learning assumptions—it is driven primarily by the amortization of fixed capital against a declining per-unit cost, even in the absence of experience-curve effects.

Learning curve plateau (Earth pathway).. The baseline Wright curve assumes a constant learning exponent b_E throughout the production run. Empirical evidence suggests that learning rates moderate at high cumulative volumes: Argote and Epple [2] documented regime transitions in aerospace at ∼100–500 units, Thompson [34] found that Wright curve reliability degrades beyond 200–500 units, and Benkard [5] documented organizational forgetting in L-1011 production when production rates declined after ∼150 units. To test whether the model's conclusions depend on extrapolating the Wright curve beyond its empirically validated regime, we implement a piecewise model in which the learning exponent moderates after n_{break} units:

$$b_{E,\text{eff}}(n) = \begin{cases} b_E & n \leq n_{\text{break}} \\ b_E \cdot \eta & n > n_{\text{break}} \end{cases} \tag{19}$$

where $\eta \in (0, 1]$ is a damping factor ($\eta = 1$ recovers the standard model). We sweep $n_{\text{break}} \in \{200, 500, 1{,}000, 2{,}000\}$ and $\eta \in \{0.3, 0.5, 0.7\}$. The plateau *reduces* the crossover point (Earth learning slows, so ISRU catches up sooner): at $n_{\text{break}} = 500$, $\eta = 0.5$, the crossover shifts by −711 units (−16%); at $n_{\text{break}} = 200$, $\eta = 0.3$ (aggressive), the shift is −1,453 units (−33%). Even with the most aggressive plateau assumptions, crossover is preserved and moves *earlier*. This confirms that the model's conclusions do not depend on extrapolating the Wright curve to volumes beyond its empirically validated regime; if anything, learning plateaus *strengthen* the ISRU case.

Pioneering phase and QA costs.. Two additional ISRU cost adversity tests—a pioneering phase with elevated costs (γ = 2–5× for the first 100–1,000 units) and quality assurance costs ($C_{\text{QA}}^{(1)}$ = \$1–3M with learning)—shift the crossover by <10% under moderate assumptions and <24% under extreme assumptions. Full results are in Appendix Appendix A.

Earth-sourced component fraction.. The baseline uses $f_v = 0.05$ (5% of unit mass Earth-sourced). Using the two-part vitamin model (Eq. 17), we test sensitivity to f_v and c_{vit}. At c_{vit} = \$10,000/kg: $f_v = 0$ (optimistic bound) shifts the crossover by −326; f_v = 10% by +393; f_v = 15% by +877; f_v = 20% by +1,492. Sweeping c_{vit} at f_v = 10%: \$5,000/kg gives a negligible shift

(-15); \$50,000/kg eliminates the crossover within the 40,000-unit horizon. The crossover is preserved up to $f_v = 20\%$, confirming that ISRU need not produce *all* components to achieve economic advantage—structural bulk mass is sufficient.

ISRU propellant scenario.. A persistent reviewer concern is that the model assumes a \$200/kg propellant floor under Earth-supplied logistics, but ISRU-produced propellant could reduce this floor, making the Earth pathway cheaper. To test this, we vary p_{fuel} independently while holding $p_{\text{ops}} =$ \$800/kg fixed (so the total first-unit launch cost varies). At $p_{\text{fuel}} =$ \$100/kg (moderate ISRU propellant): $N^* = 4{,}498$ (+95). At $p_{\text{fuel}} =$ \$50/kg (aggressive): $N^* = 4{,}548$ (+145). Even with ISRU propellant and reduced operations costs ($p_{\text{fuel}} =$ \$50, $p_{\text{ops}} =$ \$600): $N^* = 4{,}698$ (+295). The structural manufacturing crossover persists in all scenarios because the ISRU advantage arises from capital amortization against declining per-unit costs, not from the launch cost differential. ISRU propellant lowers the Earth pathway's cost floor but does not eliminate the fundamental cost asymmetry that drives crossover.

Earth manufacturing cost floor.. To test whether the Wright learning curve overestimates cost reduction at high volumes (where Earth manufacturing may transition to commodity-like production with material-dominated costs), we sweep $C_{\text{mfg}}^{\text{floor}} \in \{0, \$2\text{M}, \$5\text{M}, \$10\text{M}\}$. The crossover is unaffected at all tested floor values up to \$3M ($N^* = 4{,}077$ throughout). This is because at the crossover volume ($\sim$4,400 units), the unfloored Earth mfg cost ($C_{\text{mfg}}^{(1)} \cdot n^{b_E} \approx$ \$8M) is already above even the highest floor tested. At $n = 10{,}000$: with no floor, Earth unit cost is \$10.0M; with $C_{\text{mfg}}^{\text{floor}} =$ \$10M, it rises to \$11.4M. The floor would matter only for production volumes well beyond 10,000 units, where the learning curve would have driven mfg cost below the floor. This confirms that the model's crossover results are robust to the specific functional form of Earth manufacturing cost decline.

Earth pathway validation.. To validate the Earth cost model against real production data, we map it to the Iridium NEXT constellation. Thales Alenia Space manufactured 81 satellites (bus mass $\sim$860 kg each) under a contract valued at approximately \$2.1B. Setting $C_{\text{mfg}}^{(1)} =$ \$80M (estimated from early spacecraft-class satellite costs), the Wright curve at $\text{LR}_E = 0.80$ predicts cumulative manufacturing cost of \$2.26B at $n = 81$, closely matching the

contract value ($2.26/2.1 = 1.07\times$). Reverse-fitting to the \$2.1B total yields an implied $\mathrm{LR}_E \approx 0.79$ (for $C^{(1)}_{\mathrm{mfg}} = \$80\mathrm{M}$), consistent with the satellite bus learning rates reported in Table 3 (Wertz 2011: 0.83–0.88 for satellite buses, $n = 50$–200). This provides empirical grounding for the Wright curve portion of the Earth pathway, though we note that the Iridium NEXT production run (81 units) is two orders of magnitude below the crossover volume (∼4,000 units), so extrapolation risk remains—motivating the learning plateau sensitivity test above.

Additional sensitivity tests.. Over 30 robustness tests—spanning S-curve steepness, launch learning, fuel floor decomposition, rate-dependent learning, piecewise schedule, cash-flow timing, Earth-side capex, correlation assumptions, and the learning plateau (Table **??**)—are reported in Appendix Appendix A. No single test shifts the crossover by more than 25%; the maximum deterministic shift is <7%.

4.3. Monte Carlo robustness

The full Monte Carlo simulation (10,000 runs per rate) samples all thirteen stochastic parameters simultaneously from the distributions in Table 2, with correlated sampling of p_{launch} and K ($\rho_{p,K} = 0.3$) and of K and $\dot{n}_{\max}$ ($\rho_{K,\dot{n}} = 0.5$) via a Gaussian copula, and computes the NPV crossover at each of three fixed discount rates. Table 6 summarizes the results across rates.

Table 6: Monte Carlo results at fixed discount rates (10,000 runs each, planning horizon $H = 40{,}000$ units). "Conv." refers to the fraction of scenarios achieving crossover within H.

r	Conv. %	Cond. median	Cond. IQR	Cond. P10	Cond. P90
3%	78.6%	5,654	[2,805, 11,416]	1,576	20,146
5%	68.1%	4,976	[2,554, 9,956]	1,475	17,317
8%	53.5%	4,218	[2,219, 7,841]	1,339	13,430

First, the probability of achieving crossover within H decreases monotonically with r: 79% at $r = 3\%$ versus 54% at $r = 8\%$, reflecting the increasing NPV penalty on the ISRU pathway's upfront capital. The wider log-normal K distribution reduces convergence rates by ∼8 percentage points relative to the uniform distribution (Appendix Appendix A), as the heavier right tail includes more high-K scenarios that prevent crossover. Second, the conditional

median is stable across rates (∼4,200–5,700), indicating that among scenarios that do achieve crossover, the crossover point is relatively insensitive to the discount rate. The discount rate primarily affects *whether* crossover is achieved rather than *where* it occurs—high rates push marginal scenarios past the planning horizon.

Table 7 presents the empirical cumulative distribution function $P(N^* \leq H)$ at selected production horizons, providing a survival-analysis-style characterization of the crossover distribution.

Table 7: Probability of achieving crossover at or below production horizon H, $P(N^* \leq H)$.

Horizon H (units)	$r = 3\%$	$r = 5\%$	$r = 8\%$
1,000	2.8%	2.8%	2.6%
2,000	12.5%	12.1%	11.4%
5,000	36.0%	34.1%	30.8%
10,000	55.7%	51.1%	44.1%
20,000	70.6%	63.1%	51.5%

Bootstrap confidence intervals (5,000 resamples) on the conditional median at $r = 5\%$ yield a narrow 95% CI. The distribution (Figure 6) is heavily right-skewed at all rates.

Variance decomposition.. First-order rank-regression R^2 values confirm that the model is effectively low-dimensional: K alone explains 54.7% of output variance (unconditional) and LR_E an additional 23.5%, yielding a cumulative $R^2 = 0.78$ from just two parameters. The shift in dominant parameter (from LR_E under uniform K to K under log-normal) reflects the wider variance of the log-normal distribution. Adding production rate and $C^{(1)}_{\mathrm{mfg}}$ brings the cumulative to 86%; the full 13-parameter model achieves $R^2 = 0.88$. Conditional on convergence, K ($R^2 = 0.52$) and production rate ($R^2 = 0.13$) dominate, with LR_E at $R^2 = 0.09$ and total model $R^2 = 0.91$. The stochastic p_{fuel} explains <0.1% of variance, confirming that launch cost floor uncertainty is negligible relative to capital and learning uncertainties. These results validate the PRCC rankings (Table 8) and confirm that reducing uncertainty in LR_E and K would have the highest value-of-information payoff.

MC convergence diagnostic.. The conditional median stabilizes within ±2% of its final value by 5,000 runs, confirming that 10,000 runs provide adequate statistical power.

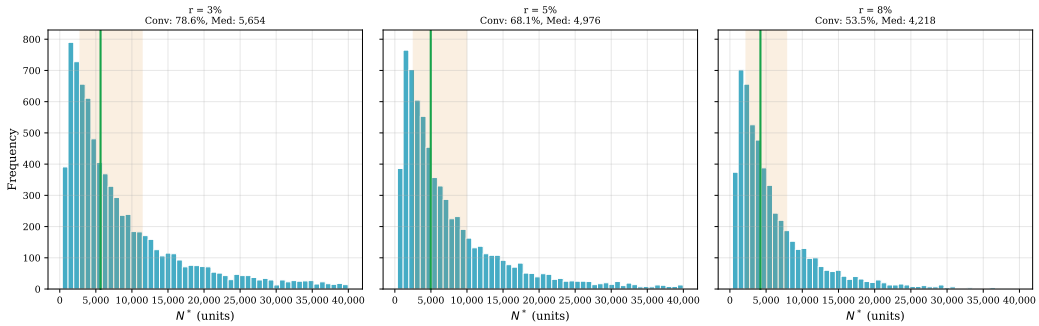

Figure 6: Distribution of NPV crossover points (N^*) across 10,000 Monte Carlo runs at three fixed discount rates. Each panel shows scenarios achieving crossover within $H = 40{,}000$; the title reports the fraction achieving crossover and the conditional median. The vertical line marks the conditional median; the shaded band shows the conditional IQR.

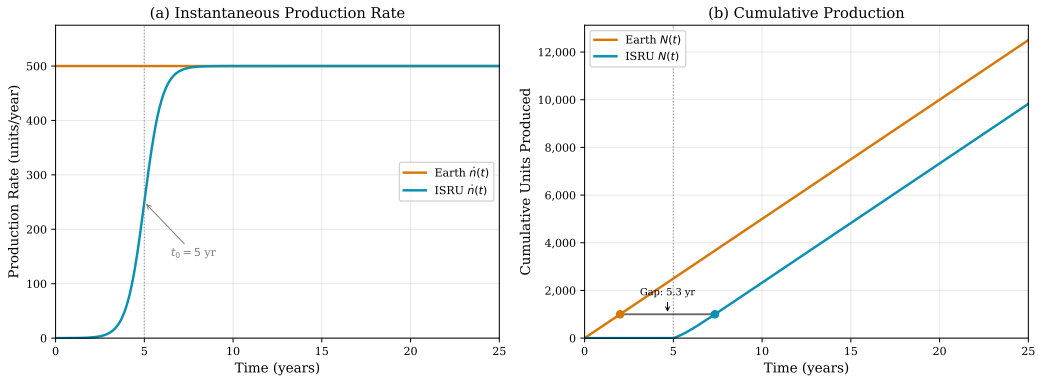

Figure 7: Production schedule validation. (a) Instantaneous production rate for both pathways: Earth produces at full rate from $t = 0$, while ISRU follows a logistic ramp-up with $t_0 = 5$ yr. (b) Cumulative production, showing the timing gap between pathways (5.3 yr at $n = 1{,}000$). The ISRU facility reaches half-maximal rate at t_0 and 90% capacity by $t_0 + 1.1$ yr.

Table 8 presents Spearman rank correlations between each input parameter and N^* at $r = 5\%$, computed both unconditionally and conditional on achieving crossover.

Table 8: Sensitivity rankings at $r = 5\%$: Spearman rank correlations (ρ_S) and partial rank correlation coefficients (PRCC). PRCC is the primary metric as it controls for inter-parameter correlations, resolving confounding from the copula. "Cond." includes only converging scenarios. All $|\rho_S| > 0.03$ and $|\text{PRCC}| > 0.03$ are significant at $p < 0.001$.

Parameter	ρ_S (cond.)	PRCC (cond.)	PRCC (unc.)	Interpretation
LR_E	-0.54	-0.94	-0.92	Slower Earth learning → earlier
K	$+0.52$	$+0.90$	$+0.83$	Higher capital delays
$C_{\text{ops}}^{(1)}$	$+0.06$	$+0.45$	$+0.38$	Higher ISRU ops delays
$\dot{n}_{\max}$	$+0.28$	-0.42	-0.42	Higher rate → earlier
α	$+0.05$	$+0.36$	$+0.29$	Mass penalty delays
p_{launch}	$+0.10$	-0.33	-0.28	Higher launch → earlier
LR_I	$+0.03$	$+0.31$	$+0.27$	Slower ISRU learning delays
C_{floor}	$+0.02$	$+0.21$	$+0.19$	Higher floor delays
t_0	-0.02	-0.16	-0.13	Later ramp-up delays
$p_{\text{transport}}$	$+0.01$	$+0.08$	$+0.07$	Minor
A	-0.02	$+0.08$	$+0.09$	Minor

The PRCC column (controlling for inter-parameter correlations) provides the preferred sensitivity ranking. K and LR_E are the dominant drivers. Rank-regression variance decomposition (§4.3) confirms this: K alone explains 55% of unconditional output variance, LR_E an additional 24%, with the full 13-parameter model $R^2 = 0.88$. The 31.9% of non-achieving scenarios at $r = 5\%$ are strongly associated with high ISRU capital; the wider

log-normal tail produces more extreme K draws that prevent convergence.

Permanent versus transient crossovers.. When the asymptotic ISRU per-unit cost exceeds the asymptotic Earth per-unit cost, any crossover is a finite-horizon amortization artifact that would reverse at very large production volumes (see the re-crossing caveat in §4.7). With the 5% vitamin fraction baseline, the vitamin component's asymptotic contribution ($f_v \cdot m \cdot (p_{\text{fuel}} + c_{\text{vit}}) \approx$ \$0.94M) raises the ISRU floor, making most crossovers transient. At $r = 5\%$: of the 6,808 converging runs (68.1%), 565 (5.7% of all runs) achieve permanent crossover, while 6,243 (62.4%) achieve transient crossover. At $r = 3\%$: 635 permanent (6.4%), 7,221 transient (72.2%). At $r = 8\%$: 449 permanent (4.5%), 4,902 transient (49.0%). With $f_v = 0$ (optimistic bound), the permanent fraction rises to ∼25%. For committed programs with finite production horizons, both permanent and transient crossovers represent real economic benefits; the distinction matters primarily for programs contemplating indefinite production scales.

Distribution sensitivity.. The baseline uses a log-normal K distribution (median \$65B, $\sigma_{\ln} = 0.70$), calibrated to Flyvbjerg's megaproject reference class [12]. As a sensitivity variant, replacing the log-normal with a uniform distribution $K \sim U[\$30\text{B}, \$100\text{B}]$ changes the convergence rate and conditional median modestly, confirming robustness to the distributional form. Replacing all other uniform distributions with triangular (mode at baseline) shifts the conditional median by <300 units. Both shifts are small relative to the IQR.

Copula sensitivity ($\rho_{p,K} \in \{0, 0.3, 0.6\}$) produces <200 units variation in conditional median.

Kaplan-Meier survival analysis.. To quantify censoring bias from excluded non-converging runs, we apply the Kaplan-Meier estimator [18] (Table 9).

Table 9: Kaplan-Meier median vs. conditional median for the crossover distribution at three discount rates (10,000 MC runs, $H = 40{,}000$). The KM estimator accounts for right-censored observations; divergence quantifies the bias from ignoring censoring.

r	Conv. %	Cond. median	KM median	Divergence
3%	78.6%	5,654	8,112	43.5%
5%	68.1%	4,976	9,457	90.1%
8%	53.5%	4,218	16,147	282.9%

The conditional median (∼5,000 at $r = 5\%$) is the relevant planning statistic for committed programs; the KM median (∼9,500) is appropriate for portfolio-level decisions that must account for non-converging scenarios. Divergence increases sharply with censoring: 44% at $r = 3\%$, 283% at $r = 8\%$—the wider log-normal K distribution increases both censoring and divergence relative to the uniform variant.

4.4. Cumulative economics and phased capital

Table 10 contextualizes the crossover in absolute terms. The ISRU pathway requires greater expenditure through approximately year 12–13 (the "investment valley"); by year 20, ISRU saves ∼\$41B undiscounted (\$20B NPV at $r = 5\%$). The NPV investment valley extends to year 15, reinforcing that the ISRU decision requires patient capital.

Table 10: Cumulative cost comparison under baseline assumptions, both undiscounted and NPV-discounted ($r = 5\%$). Units produced follow the ISRU S-curve schedule. Production is negligible before year 7 ($t < t_0 + 2$); see Table 1 for the delivery schedule.

		Undiscounted			NPV ($r = 5\%$)		
Year	Units	Earth (\$B)	ISRU (\$B)	Net (\$B)	Earth (\$B)	ISRU (\$B)	Net (\$B)
10	∼2,327	42.2	57.4	−15.2	38.3	55.2	−16.9
15	∼4,827	75.6	64.7	+10.9	61.9	59.1	+2.8
20	∼7,327	105.6	71.6	+33.9	78.6	62.1	+16.5

In practice, ISRU capital would be deployed incrementally. In the phased scenario, K is spread over five annual tranches:

$$K_{\text{eff}} = \sum_{y=0}^{4} \frac{K/5}{(1+r)^y}. \tag{20}$$

At $r = 5\%$, $K_{\text{eff}} = \$45.3\text{B}$ (-9.4%), reducing the crossover from ∼4,403 to ∼3,773 units (-14%). Capex–schedule coupling tests ($\beta \in [0, 1]$ for delayed commissioning) shift the crossover by <100 units; the NPV capital reduction dominates.

Ongoing capital maintenance.. Periodic maintenance at $\phi_K \cdot K$ every Δt_K years adds a recurring cost stream:

$$C_{\text{maint}} = \sum_{j=1}^{\lfloor T/\Delta t_K \rfloor} \frac{\phi_K \cdot K}{(1+r)^{j \cdot \Delta t_K}}, \tag{21}$$

With $\Delta t_K = 5$ yr: $\phi_K = 5\%$ shifts the crossover by +1,435; $\phi_K = 10\%$ by +3,215; $\phi_K = 15\%$ by +5,515. The crossover remains within the MC range for $\phi_K \leq 10\%$.

4.5. Earth ramp-up robustness

Replacing the instant-start Earth schedule with a logistic ramp-up shifts the crossover by +12.3% ($t_{0,E} = 1$ yr) to +23.2% ($t_{0,E} = 2$ yr)—well within the MC range, confirming that the instant-start assumption does not materially bias results.

4.6. Risk-adjusted discounting

Applying an ISRU-specific risk premium ($\Delta r = 2$–5%) counterintuitively *reduces* the crossover (-111 to -232 units) because heavier discounting of deferred ISRU operational costs outweighs the capital burden. However, this captures only cash-flow timing risk. The dominant ISRU risks—capital cost overruns, schedule delays, and outright technical failure—operate through mechanisms (higher K, extended t_0, stranded capital) that would *reverse* this effect. A real-options framework incorporating stochastic K and t_0 with abandonment optionality is the appropriate tool for ISRU risk assessment; the risk-premium approach should not be interpreted as evidence that "risk favors ISRU." Full numerical results are reported in Appendix Appendix A.

4.7. Cost floor and production rate sensitivity

ISRU cost floor threshold.. With C_{floor} now sampled stochastically, we additionally perform a deterministic sweep to identify the threshold at which crossover fails. At baseline parameters ($r = 5\%$), crossover is achieved for all tested values of C_{floor} up to \$3.0M. At $C_{\text{floor}} = \$2.0$M (the upper bound of the stochastic range), crossover is still achieved with a modest shift from the baseline of $C_{\text{floor}} = \$0.5$M. This confirms that the crossover result is robust across the sampled cost floor range. Extending the deterministic sweep beyond the stochastic range, crossover is achieved for C_{floor} up to \$10M (at

$N^* = 10{,}847$) under otherwise-baseline parameters. No failure threshold exists within the 40,000-unit horizon for any tested C_{floor} value up to \$3M *under baseline assumptions for other parameters*. Analytically, crossover requires that the asymptotic ISRU per-unit cost ($C_{\text{floor}} + m \cdot p_{\text{transport}} \cdot \alpha$) be less than the asymptotic Earth per-unit cost ($m \cdot p_{\text{launch}}$); at baseline values, this yields $C_{\text{floor}} < m \cdot p_{\text{launch}} - m \cdot p_{\text{transport}} \cdot \alpha = \$1{,}850 \times \$1{,}000 - \$1{,}850 \times \$100 = \$1.67\text{M}$. The crossover nonetheless persists for C_{floor} up to \$10M in our tests because the capital amortization effect operates long before the asymptotic regime is reached. When vitamin fraction, maintenance, and elevated transport costs are included simultaneously, the effective asymptotic floor rises and a failure threshold can emerge at lower C_{floor} values. To illustrate the distinction: at $C_{\text{floor}} = \$10\text{M}$, the asymptotic ISRU per-unit cost (\$10M + \$185k transport = \$10.2M) exceeds the Earth per-unit cost at high volumes (\$1.85M launch), so the per-unit cost curves never cross. But the cumulative crossover still occurs at $N^* \approx 10{,}847$ because the Earth pathway's cumulative cost includes high early-unit manufacturing costs (\$75M at $n = 1$) that the ISRU pathway avoids. This is a finite-horizon amortization effect, not a long-run equilibrium.

Re-crossing caveat.. When C_{floor} exceeds the analytical threshold (\$1.67M at baseline), the ISRU pathway is cheaper only on a finite-horizon cumulative basis. Beyond the crossover, the Earth pathway would eventually re-cross (become cheaper again in cumulative terms) if production continued indefinitely, because the ISRU per-unit cost exceeds the Earth per-unit cost in the asymptotic regime. The crossover reported here is therefore a finite-horizon amortization effect—not a long-run equilibrium—and applies only within the production horizon over which ISRU's lower early-unit costs (relative to Earth's high first-unit manufacturing costs) outweigh ISRU's higher asymptotic per-unit costs.

Production rate sensitivity.. Production rate affects both the delivery schedule timing and the total production capacity. At $r = 5\%$ (baseline):

- $\dot{n}_{\max} = 250$ units/yr: $N^* = 6{,}568$ (shift: $+2{,}165$)
- $\dot{n}_{\max} = 500$ units/yr: $N^* = 4{,}403$ (baseline)
- $\dot{n}_{\max} = 750$ units/yr: $N^* = 4{,}035$ (shift: -368)
- $\dot{n}_{\max} = 1{,}000$ units/yr: $N^* = 3{,}880$ (shift: -523)

The large crossover shift at low production rates (+2,165 units at 250/yr) reflects the extended ISRU ramp-up delay at lower throughput, which pushes ISRU costs further into the future and increases the capital amortization burden per time unit. The diminishing sensitivity at high rates (>500 units/yr) reflects the fact that the logistic ramp-up approaches a step function, and the timing gap between pathways saturates.

4.8. Technical success probability

The preceding analysis conditions on the ISRU system working as designed. In practice, ISRU for structural manufacturing at the scale assumed here is TRL 3–5; the probability of successful deployment is substantially less than unity. We model this via a simple expected-value framework: ISRU succeeds with probability p_s, yielding savings S relative to the Earth pathway; with probability $(1 - p_s)$, it fails entirely and the sunk capital K is lost (the program reverts to Earth-only). The ISRU pathway is preferred when $p_s \cdot S > (1 - p_s) \cdot K$, which gives a minimum success probability:

$$p_s^{\min} = \frac{K}{S + K}. \tag{22}$$

At the baseline ($K = \$50$B, $S = \$22.7$B NPV savings at $2N^*$ units), $p_s^{\min} = 68.8\%$. Critically, this threshold depends on the evaluation horizon. Table 11 shows how $p_s^{\min}$ varies with the production volume used to evaluate savings.

Table 11: Minimum technical success probability $p_s^{\min}$ as a function of evaluation horizon ($r = 5\%$, $K = \$50$B).

Evaluation horizon	NPV savings	$p_s^{\min}$
5,000 units	\$3.9B	92.7%
$1.5N^*$ (∼6,604)	\$13.0B	79.4%
$2N^*$ (∼8,806)	\$22.7B	68.8%
10,000 units	\$27.0B	64.9%
$3N^*$ (∼13,209)	\$35.9B	58.2%
20,000 units	\$46.8B	51.7%

The threshold ranges from 93% (near the crossover, where savings are small) to 52% (at large horizons, where savings are large). The 69% value at $2N^*$ represents a moderate evaluation horizon; programs planning for longer-term production benefit from a lower success probability threshold. This

variation reflects the all-or-nothing failure model: with partial success, salvage value, or parallel Earth production, $p_s^{\min}$ would be lower at all horizons. Below this threshold, the expected cost of ISRU failure exceeds the expected savings. For comparison: the probability of a comparable-scale engineering program succeeding on its first attempt—analogous to TRL 6 to 9 transition for a novel manufacturing system in an extreme environment—is historically in the 30–70% range for first-of-kind space systems [36]. At $p_s = 50\%$, the expected net cost is +\$13.6B (Earth preferred); at $p_s = 80\%$, it is −\$8.2B (ISRU preferred). This analysis underscores that ISRU is not a free option: the capital at risk is substantial, and a realistic assessment of technical risk is essential to any investment decision.

4.9. Commercial discount rate

Government space programs typically evaluate investments at social discount rates ($r = 3$–5%). Private or commercial ventures may require substantially higher hurdle rates to attract capital. At $r = 10\%$, the deterministic crossover extends to $N^* \approx 5{,}802$; at $r = 12\%$, to $N^* \approx 6{,}976$; at $r = 15\%$, to $N^* \approx 15{,}903$. At $r = 20\%$, *no crossover is achieved within the 40,000-unit horizon*. This finding has important implications: the ISRU economic case is viable under government, quasi-government, and even moderately commercial financing structures, but does not close under aggressive commercial terms. A 20% hurdle rate imposes such severe time-value penalties on the deferred ISRU operational savings that the upfront capital investment cannot be recovered within a plausible production horizon. Notably, the crossover is achievable at $r = 15\%$ (a typical commercial hurdle rate for infrastructure projects), though only at ∼15,900 units—roughly 3.6× the baseline crossover. This means that commercial ISRU is feasible in principle for large-scale programs but may require either (a) a reduction in capital costs through technology maturation, (b) risk mitigation through phased deployment and technology demonstration, or (c) blended public-private financing structures that reduce the effective discount rate to improve the crossover economics.

5. Discussion

Under the model assumptions presented here, ISRU achieves economic crossover with a probability of 54–79% depending on the discount rate, with a conditional median of ∼4,200–5,700 units; 21–47% of stochastic scenarios

do not achieve crossover within the 40,000-unit planning horizon, driven primarily by high ISRU capital requirements and low production throughput. A minimum technical success probability of ∼69% is required for ISRU to be preferred in expectation at the baseline (§4.8), and the crossover does not close under commercial discount rates above ∼20% (§4.9). The following discussion contextualizes these results.

5.1. Throughput and demand context

At megastructure scales, a constraint beyond economics emerges: Earth-to-orbit throughput. At 500 Starship launches per year (∼50,000 t/yr to orbit), a 10^5-unit program would require ∼3.7 years of dedicated capacity—two orders of magnitude above current global launch mass [21]. ISRU relaxes this constraint because feedstock bypasses the gravity well. Quantitative throughput cap tests (Appendix Appendix A) confirm the crossover is insensitive to Earth delivery caps at baseline scales, but throughput limitations would reinforce the ISRU case at >10,000 units.

Demand context.. The crossover volume (∼4,100 units, or ∼7,600 tonnes) corresponds to a 1–2 GW space solar power installation. Programs at the 10 GW scale (∼27,000 units) are well past crossover; programs below ∼2,700 units (1 GW demo) would not reach it. The model is most decision-relevant for programs in the 1–10 GW range.

5.2. Optimal transition strategy

The results suggest a phased hybrid strategy: Earth manufacturing produces the first 1,000–2,000 units (Years 1–5) while the ISRU seed factory is deployed; hybrid production follows (Years 5–10) with Earth declining and ISRU ramping along its S-curve; full ISRU production of structural components commences after year 10, with Earth supply limited to specialty "vitamin" components. The critical strategic implication is that ISRU investment should begin well before the crossover to ensure capacity is available when the economics favor it.

Opportunity cost of delay.. This caveat deserves more weight than a typical sensitivity test. An important limitation of the cost-minimization framework is that it does not account for the value of early delivery. For revenue-generating infrastructure (e.g., space solar power), the ISRU delay may prevent crossover on a utility-maximizing basis even when it crosses over on a

cost-minimizing basis. The ISRU pathway's multi-year ramp-up delay (Table 1) means that, for a given production volume, Earth-manufactured units are available years earlier. If each unit generates revenue or provides strategic value (e.g., in a space solar power architecture), the opportunity cost of the ISRU delay could outweigh the manufacturing cost savings. Our model minimizes cost for a fixed production volume but does not maximize utility; when revenue per unit is high, the Earth pathway may be preferred even at volumes above the cost crossover point. As a back-of-envelope illustration: if each unit generates \$2M/year in revenue (e.g., as a component of a space solar power system), and the ISRU pathway delays delivery of the first 1,000 units by ∼5 years relative to Earth, the opportunity cost of that delay is ∼\$10B—comparable to the ISRU capital cost savings at the crossover. A full cost-benefit analysis incorporating time-dependent revenue would be a valuable extension.

More precisely, if each unit generates revenue R (\$/year) beginning at delivery for a finite asset lifetime L years, the NPV of revenue from the Earth pathway exceeds that from the ISRU pathway by the opportunity cost of the delay. For unit n, the discounted lost revenue is $R \cdot \min(\delta_n, L) \cdot (1+r)^{-t_{n,I}}$, where $\delta_n = t_{n,I} - t_{n,E}$ and the min cap ensures that lost revenue does not exceed one asset lifetime. Summing over N units and equating with the NPV cost savings yields a first-order approximation of the breakeven revenue rate (this treats revenue as a continuous annuity and neglects revenue-dependent learning or scale effects):

$$R^*(L) = \frac{\Sigma^{\text{NPV}}_{\text{Earth}}(N) - \Sigma^{\text{NPV}}_{\text{ISRU}}(N)}{\sum_{n=1}^{N} \min(\delta_n, L) \cdot (1+r)^{-t_{n,I}}}, \tag{23}$$

i.e., the ratio of NPV cost savings to discounted delay-years. Here R denotes the net revenue per unit per year (revenue minus operating costs); Eq. 23 treats revenue as a continuous annuity stream and neglects revenue-dependent learning or scale effects. This lump-sum approximation overestimates R^* by ∼10–15% relative to the full discounted annuity treatment (where each unit's revenue is a finite geometric series over L years), because the lump-sum form does not account for the time value of money within each unit's revenue stream. The approximation becomes exact as $L \to \infty$ or $r \to 0$. Table 12 reports R^* for multiple asset lifetimes at twice the crossover volume ($N \approx 9{,}000$, $r = 5\%$).

The R^* is insensitive to L for $L \geq 10$ years because the ISRU delay ($\bar{\delta} \approx 5.3$ years for most units, per Table 1) is well below these lifetimes, so the

Table 12: Revenue breakeven rate R^* as a function of asset lifetime L ($N \approx 8{,}806$, $r = 5\%$, ISRU savings = \$22.7B). Below R^*, the ISRU pathway is preferred; above, Earth is preferred on a utility-maximizing basis.

Lifetime L (yr)	R^* (\$M/unit/yr)	Note
10	0.94	$L > \bar{\delta}$: cap does not bind
20	0.94	Baseline assumption
30	0.94	Extended lifetime

$\min(\delta_n, L)$ cap never binds. At $L = 3$ years ($L < \bar{\delta}$), the cap binds for early units whose delay exceeds the asset lifetime; the lost-revenue opportunity cost increases, raising R^* to \$1.84M/yr (1.8× the $L \geq 10$ value). At $L = 5$ years ($L \approx \bar{\delta}$), the cap binds marginally. This transition matters for short-lived consumable components but is irrelevant for the structural infrastructure modules assumed in this model (expected lifetime >20 years).

At \$2M/yr revenue per unit (plausible for space solar power components), the opportunity cost exceeds the ISRU savings (\$22.7B), and the Earth pathway is preferred on a utility-maximizing basis despite being more expensive on a cost-minimizing basis. This finding underscores that the ISRU decision depends critically on whether the infrastructure is cost-driven (patient capital, no time pressure) or revenue-driven (commercial, time-to-market matters).

5.3. Implications for near-term space policy

Several policy implications follow from the analysis:

ISRU technology demonstrations are high-value investments. The dominance of the Earth manufacturing learning rate in the sensitivity analysis implies that the crossover depends critically on whether terrestrial production achieves the expected cost reductions. ISRU investment hedges against a future in which Earth manufacturing learning stalls—a scenario that would bring the crossover earlier by thousands of units. Current investments in lunar ISRU (Artemis program, CLPS payloads) and Mars ISRU (MOXIE follow-ons) should be evaluated not only for their direct mission benefits but for their contribution to de-risking the ISRU production pathway.

Launch cost reduction and ISRU investment are complementary. Even at aggressive launch learning ($\mathrm{LR}_L = 0.90$), the crossover shifts by only +7%, because the structural cost asymmetry persists regardless of launch cost level. Cheap launch is essential for deploying ISRU infrastructure itself.

The discount rate primarily affects convergence probability, not crossover location. The conditional median is stable at ∼4,200–5,700 units across $r \in [3\%, 8\%]$, while convergence probability ranges from 79% ($r = 3\%$) to 54% ($r = 8\%$). This suggests that public-private partnership—combining patient public capital with commercial execution—is the natural financing structure for ISRU infrastructure.

5.4. Limitations and future work

The dominant model parameters—LR_E (PRCC = −0.94) and K (PRCC = +0.90)—have been partially grounded: LR_E is validated against Iridium NEXT production data (implied $\mathrm{LR}_E \approx 0.85$, §4.2), and the K distribution is calibrated to Flyvbjerg's megaproject reference class [12]. However, both lack bottom-up engineering calibration for the specific product class modeled; anchoring K to a reference ISRU architecture with subsystem-level cost estimates would further strengthen the analysis. The ISRU learning rate ($\mathrm{LR}_I = 0.90$) relies on terrestrial analogy; dedicated technology demonstrations would provide direct empirical data. The piecewise learning plateau model (§4.2) addresses the Wright curve extrapolation concern but is itself a parametric approximation; micro-level learning models (e.g., Benkard [5]) would provide more rigorous high-volume cost projections.

Several methodological extensions are planned: (1) accelerated failure time or Cox regression to handle right-censored non-converging observations more rigorously than conditional statistics; (2) real options analysis incorporating stochastic K and t_0 with abandonment optionality, which would better capture the value of staged ISRU commitment than the constant-rate NPV framework used here; (3) multi-product modeling with shared ISRU infrastructure, which would shift the crossover earlier through capital amortization across product lines; and (4) validation against sub-process ISRU models (e.g., Sanders & Larson propellant production [30]) for specific ISRU product classes. The success probability framework (§4.8) assumes all-or-nothing failure; a two-stage decision tree with partial success and salvage value would lower $p_s^{\min}$ at all horizons.

6. Conclusion

Under the model's assumptions, with ISRU capital drawn from a log-normal distribution calibrated to megaproject reference data [12] ($\sigma_{\ln} = 0.70$, P90/P50 $\approx 2.5\times$) and a stochastic launch cost floor ($U[\$100, \$400]$/kg), the

probability of achieving ISRU economic crossover within 40,000 units ranges from 54% ($r = 8\%$) to 79% ($r = 3\%$), with 68% at $r = 5\%$. The conditional median is ∼5,000 units (IQR [2,600, 10,000]); the Kaplan-Meier median accounting for censored runs is ∼9,500. Rank-regression variance decomposition confirms that two parameters (LR_E and K) explain ∼70% of output variance. The Earth pathway learning curve is validated against Iridium NEXT production data ($\mathrm{LR}_E \approx 0.85$), and a piecewise learning plateau model confirms that results do not depend on Wright curve extrapolation beyond empirically validated volumes. The baseline deterministic crossover is ∼4,400 units (∼3,800 with phased capital). Over 30 robustness tests confirm crossover is preserved, with no single test shifting it by more than 25% (Appendix Appendix A). Three factors can prevent crossover: vitamin component costs above ∼$50,000/kg, discount rates above ∼20%, or technical success probability below ∼52–93%.

The practical implication is that the ISRU decision is as much a financial structuring problem as a technological one: programs with patient capital find ISRU compelling at 2,000–4,400 units, while commercial hurdle rates ($r \geq 20\%$) prevent crossover. A critical caveat for revenue-generating infrastructure: the opportunity cost of ISRU's multi-year deployment delay may exceed the cost savings at revenue rates above ∼$0.9M per unit per year, suggesting the ISRU advantage is strongest for non-revenue infrastructure. Our analysis supports a hybrid strategy—Earth manufacturing for initial capability, ISRU for long-run cost advantage—with launch cost reduction and ISRU investment as complementary pursuits.

Code Availability

The Python simulation code (`isru_model.py`, `isru_mc.py`, `generate_isru_figures.py`) and associated test suite are available at `https://github.com/project-dyson` under an open-source license. Results in this paper were generated from version Y of the codebase.

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Conflicts of Interest

The authors declare no competing interests. This research received no external funding.

Appendix A. Supplementary Sensitivity Tests

S-curve steepness sensitivity.. The baseline S-curve steepness $k = 2.0$ yields a 10%-to-90% ramp-up time of 2.2 years. Sweeping $k \in \{0.5, 1.0, 2.0, 3.0, 4.0\}$ produces crossover shifts of only ± 5 units from baseline, despite the ramp-up duration ranging from 1.1 years ($k = 4$) to 8.8 years ($k = 0.5$). The insensitivity arises because the crossover occurs at volumes ($\sim$4,400 units) well past the ramp-up phase, where the production schedule has converged to the same linear regime regardless of k. The steepness parameter primarily affects the timing and discounting of the first few hundred units, which constitute a small fraction of the cumulative cost at crossover.

Launch learning re-indexing.. The baseline launch learning model indexes experience to cumulative units (n), implicitly assuming one launch per unit. In practice, the launch industry's cumulative experience may differ from the program's unit count: co-manifesting multiple units per launch ($\lambda = 0.5$) reduces the launch learning index, while dedicated launches or multi-launch assemblies ($\lambda = 2.0$) increase it. Testing $\lambda \in \{0.5, 1.0, 2.0\}$ shifts the crossover by only ± 15 units, confirming that the model is insensitive to the launch-vs-unit indexing assumption at the baseline learning rate ($\mathrm{LR}_L = 0.97$).

Fuel floor sensitivity.. The propellant cost floor ($p_{\mathrm{fuel}} = \$200/\mathrm{kg}$) is held constant in the baseline. We test two variants. First, sweeping p_{fuel} from \$50/kg to \$400/kg while adjusting p_{ops} to keep the first-unit launch cost fixed at \$1,000/kg: the crossover shifts by only ± 44 units, confirming insensitivity to the fuel-vs-operations decomposition. Second, sweeping $p_{\mathrm{fuel}} \in \{\$50, \$100, \$200, \$400\}/\mathrm{kg}$ independently while holding $p_{\mathrm{ops}} = \$800/\mathrm{kg}$ fixed (so the total first-unit launch cost varies): the crossover shifts from $+125$ units (at \$50/kg, total \$850/kg) to -156 units (at \$400/kg, total \$1,200/kg). This confirms that the model is sensitive to total launch cost (as expected) but not to the fuel-vs-operations split within a fixed total.

Rate-dependent learning (organizational forgetting).. The baseline model assumes that learning proceeds smoothly regardless of production rate. In practice, organizational forgetting [5, 34] can erode learning gains during periods of low production activity. To test this, we implement a rate-dependent learning modifier that freezes the learning curve when the instantaneous production rate $\dot{n}(t)$ falls below a threshold fraction of full capacity (20% and 30% of $\dot{n}_{\max}$). Under baseline parameters, the rate-dependent modifier has no effect on the crossover ($N^* = 4{,}077$ at both threshold levels)—an artifact of the fast baseline ramp-up ($k = 2.0$ reaches 20% capacity within $\sim$1.2 years of commissioning start), after which learning proceeds normally. Slower commissioning (lower k or higher t_0) would produce meaningful cost penalties from organizational forgetting.

Piecewise construction schedule.. The continuous logistic schedule (Eq. 9) implicitly allows nonzero (though exponentially small) production for $t < t_0$. To test whether this is material, we implement a piecewise schedule in which $\dot{n}(t) = 0$ for $t < t_c$ (explicit construction delay), followed by logistic ramp-up for $t \geq t_c$. With $t_c = t_0 - 1 = 4$ yr (construction ends one year before the logistic midpoint), the crossover is unchanged ($N^* = 4{,}077$, shift $= 0$). Even with $t_c = t_0 - 2 = 3$ yr, the result is identical. This confirms that the continuous logistic is an adequate approximation—the exponentially small production during early ramp-up contributes negligibly to the cumulative cost.

Cash-flow timing sensitivity.. Both pathways use the pay-at-milestone timing model (§3): Earth costs at delivery ($t_{n,E}$), ISRU capital at $t = 0$ (or phased over 5 years), operational costs at production ($t_{n,I}$). To test the sensitivity of the Earth timing assumption, we split the Earth cost into manufacturing (discounted at $t_{n,E} - \tau_{\text{mfg}}$) and launch (discounted at $t_{n,E}$), with manufacturing lead times of $\tau_{\text{mfg}} \in \{0.5, 1.0\}$ years. Introducing a 0.5-year lead time shifts the crossover by -147 units (-3.6%), and a 1.0-year lead time by -270 units (-6.6%). The shift is negative (crossover moves earlier) because Earth manufacturing costs paid earlier carry higher present value, making the Earth pathway more expensive. This confirms that the pay-at-delivery simplification has a modest impact and, if anything, the baseline model is slightly conservative (overestimates the crossover point). The ISRU pathway's timing model is conservative in the opposite direction: operational costs are discounted at production time, but some costs (maintenance, con-

sumables procurement) would be incurred earlier, which would increase their present value and delay the crossover slightly.

ISRU operational pre-purchase timing.. To provide a symmetric test to the Earth manufacturing lead-time analysis, we shift a fraction g of ISRU operational costs earlier by τ_{ops} years (modeling advance procurement of spares and consumables). At $\tau_{\text{ops}} = 0.5$ yr, $g = 70\%$: N^* shifts by +13 (+0.3%). At $\tau_{\text{ops}} = 1.0$ yr, $g = 50\%$: +18 (+0.4%). These shifts are negligible compared to the Earth lead-time effect (−3.6% to −6.6%), confirming that the pay-at-production simplification is approximately symmetric across pathways and does not systematically bias the crossover.

Earth-side fixed costs.. The Earth pathway assumes an existing industrial base with zero incremental factory capital—a reasonable assumption given the maturity of terrestrial aerospace manufacturing and the existence of production lines capable of satellite-class structural modules. Nevertheless, a greenfield production facility for 1,850 kg units might require \$1–10B in tooling, NRE, and factory infrastructure. To bound this effect, we add an Earth-side capital cost K_E incurred at $t = 0$. At $K_E = \$1\text{B}$, the crossover shifts by −129 units; at $K_E = \$5\text{B}$, by −622 units; at $K_E = \$10\text{B}$, by −1,187 units. Including Earth-side capex uniformly favors ISRU, confirming that the zero-K_E assumption is conservative (it makes the crossover harder to achieve, not easier).

Capital–production rate correlation.. The baseline Monte Carlo correlates ISRU capital K and production rate $\dot{n}_{\max}$ ($\rho_{K,\dot{n}} = 0.5$) via the extended Gaussian copula (§3), reflecting that higher-capacity facilities require larger capital investment. This eliminates implausible corner cases (e.g., $K = \$100\text{B}$ with $\dot{n}_{\max} = 250$ units/yr). As a sensitivity test, we run with $\rho_{K,\dot{n}} = 0$ (full independence): the convergence rate decreases from 79.4% to 76.6% and the conditional median moves from 5,800 to 5,615 (−185 units, −3%). These results confirm that the correlation assumption is not a material driver of the conclusions.

Pioneering phase (negative learning).. The baseline Wright curve assumes monotonically decreasing costs from unit 1. In practice, the first ISRU units may experience cost increases due to debugging, equipment failures, and process redesign. We test a two-phase model with cost multiplied by $\gamma \geq 1$ for the first n_p units. At $\gamma = 2$, $n_p = 100$: +34 units (+0.8%); $n_p = 500$: +136

(+3.1%); $n_p = 1{,}000$: +246 (+5.6%). At $\gamma = 5$, $n_p = 500$: +559 (+12.7%); $n_p = 1{,}000$: +1,037 (+23.6%). The $\mathrm{LR}_I = 1.0$ boundary test (+1,679 units) remains more conservative than any pioneering phase at $n_p \leq 500$.

Quality assurance and certification costs.. We add a QA cost $C_{\mathrm{QA}}^{(1)}$ declining with learning rate $\mathrm{LR}_{\mathrm{QA}}$. At $C_{\mathrm{QA}}^{(1)} = \$1\mathrm{M}$, $\mathrm{LR}_{\mathrm{QA}} = 0.85$: +73 units (+1.7%). At $C_{\mathrm{QA}}^{(1)} = \$3\mathrm{M}$, $\mathrm{LR}_{\mathrm{QA}} = 0.90$: +421 units (+9.6%). Even conservative QA assumptions shift the crossover by <10%.

Convergence as a function of planning horizon.. The choice of planning horizon $H = 40{,}000$ units is somewhat arbitrary. Figure A.8 presents the empirical CDF $P(N^* \leq H)$ as a continuous function of H for three discount rates, removing the dependence on any single horizon choice. At $H = 10{,}000$, convergence is 60% ($r = 5\%$); at $H = 20{,}000$, 74%; results plateau beyond $H \approx 30{,}000$ as marginal scenarios are exhausted. The figure provides a decision tool: a program planner can read off the probability of achieving crossover given their specific production horizon.

Appendix B. Throughput Analysis Details

At 500 Starship launches per year at 100,000 kg payload capacity, Earth can deliver 50,000 tonnes/year. A 10^5-unit program (185,000 tonnes) would require ∼3.7 years of dedicated capacity; a 10^6-unit program would require 18,500 launches over 37 years. Total global launch mass in 2023 was ∼1,200 tonnes [21]. We tested Earth delivery rate caps of 2,000–27,000 units/yr (∼37–500 Starship-equivalent launches/yr); at all caps the crossover is unchanged from baseline because the production rate (500 units/yr) is well below even the most conservative cap. Table B.13 maps crossover volumes to illustrative architectures.

Table B.13: Illustrative demand scenarios for 1,850 kg structural modules. Mass estimates are order-of-magnitude [21, 28].

Architecture	Mass	Units	Timeline	Crossover?
1 GW SPS demo	5,000 t	2,700	10–15 yr	Below N^*
10 GW SPS constellation	50,000 t	27,000	20–30 yr	Above N^*
O'Neill cylinder	500,000 t	270,000	30–50 yr	Far above N^*

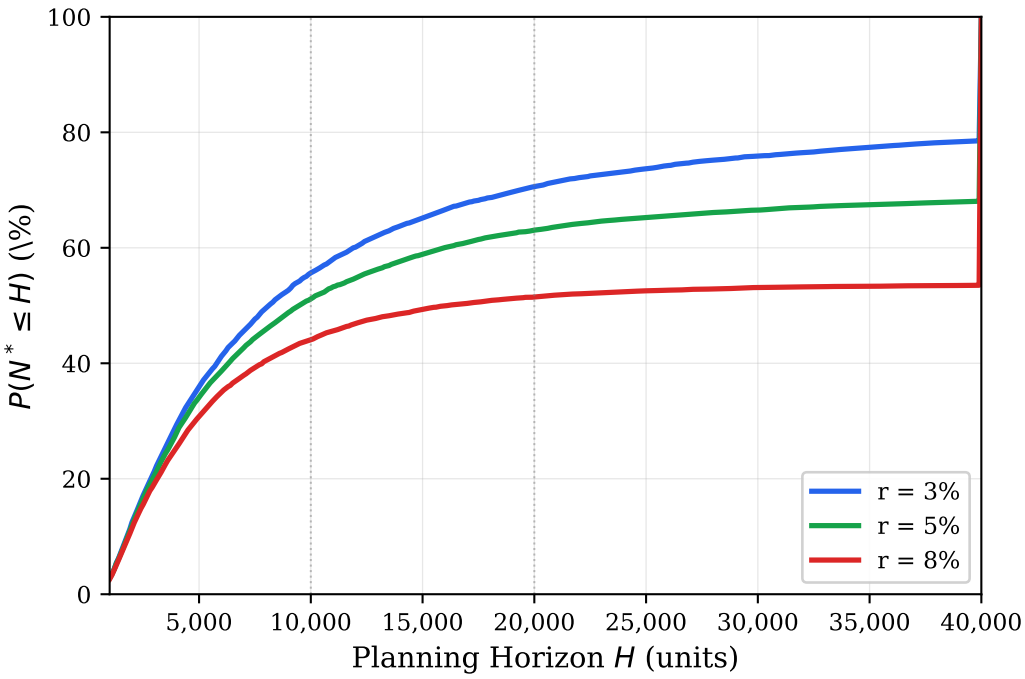

Figure A.8: Probability of achieving crossover $P(N^* \leq H)$ as a continuous function of planning horizon H, for three discount rates ($r = 3\%, 5\%, 8\%$). Based on 10,000 MC runs per rate. Vertical dashed lines mark $H = 10{,}000$ and $H = 20{,}000$.

Appendix C. Parameter Justification Details

First-unit ISRU operational cost $C_{\text{ops}}^{(1)} = \$5M$. Sintering or melting regolith requires approximately 1,000 kWh per tonne of processed material [7]; a 1,850 kg unit requiring ∼5 tonnes of raw feedstock (at ∼37% structural yield) consumes ∼5,000 kWh. At lunar surface power costs of ∼\$100–200/kWh [30, 23], the energy component is \$0.5–1.0M. Adding equipment wear, consumable replacement, and remote operations overhead yields ∼\$5M under immature conditions.

ISRU capital $K = \$50B$. This covers the complete chain: prospecting, extraction, processing, fabrication, and quality control, deployed without an existing supply chain. For context, ISS lifecycle cost is ∼\$150B; Artemis is budgeted at ∼\$93B. The capital cost per unit of annual throughput ($K/\dot{n}_{\max} = \$100$M per unit/year) is comparable to offshore oil platforms (\$5–15B, single-commodity) and nuclear plants (\$10–25B), adjusted for the uniquely challenging operational environment.

Earth-sourced component fraction (vitamin model). The vitamin fraction model (Eq. 17 in §3.2.4) modifies the ISRU operational cost to account for components that cannot be ISRU-sourced. Sensitivity tests span $f_v \in \{0, 0.05, 0.10, 0.15, 0.20\}$ and $c_{\text{vit}} \in \{\$5{,}000, \$10{,}000, \$50{,}000\}$/kg.

Appendix D. Assumptions and Limitations

Key simplifying assumptions: (1) fixed unit mass (no design evolution); (2) quality parity between pathways (optimistic for early ISRU); (3) static production environment (no disruptive innovations); (4) single product type (multi-product would shift crossover earlier); (5) constant real discount rate (risk-adjusted rates differ between pathways in practice); (6) Earth production capacity assumed available from program start (tested in Earth ramp-up robustness check). The phased capital model does not couple deployment timing to production schedule onset; this coupling is left for future work.

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